

Anisotropic Non-Negative Kernel Control on the Oriented Frame Bundle of the Sphere: Holonomy, Sequence Dependence, and the WGS84 Lollipop Problem

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Abstract

We study the problem of approximating a prescribed non-negative radial-removal field on the unit two-sphere $\mathbb{S}^2$ by a finite superposition of localised anisotropic Gaussian kernels, each parameterised by an oriented tangent frame $R_n \in SO(3)$, a non-negative strength $\alpha_n \geq 0$, and a positive-definite tangent-plane covariance $\Sigma_n \in \text{Sym}^+(2)$. The motivating application is the WGS84 lollipop problem: shaping a spherical candy into a miniature oblate spheroid matching the WGS84 geodetic reference ellipsoid by a sequence of localised licking operations. The target removal field $\delta_{\text{target}}(\theta) = R_0 f \cos^2 \theta$ is shown to lie entirely in the $\ell = 0$ and $\ell = 2$ spherical-harmonic modes, making it spectrally sparse.

The central contribution is the identification of the contact state of an anisotropic surface-removal operation with an element of $SO(3)$, using the fibration $\mathbb{S}^2 \cong SO(3)/SO(2)$. Because the Levi-Civita connection on $\mathbb{S}^2$ has non-trivial holonomy proportional to enclosed solid angle, parallel transport of a tangent frame around a non-geodesic closed loop generates a rotation of the tangent basis by the enclosed area (Gauss–Bonnet). Consequently, two sequences visiting the same contact points in different orders may accumulate different total removal fields in the anisotropic case. This makes the optimal sequencing of anisotropic surface operations a control problem on $SO(3)$, not merely a placement problem on $\mathbb{S}^2$.

We prove rigorously that the isotropic model is the $SO(2)$ -symmetry-reduced limit of the frame-aware model; that holonomy-induced orientation rotation constitutes a measurable mismatch in the anisotropic case; and that sequence-order dependence disappears in the isotropic limit. A discrete optimisation problem on $(SO(3))^N$ and a continuous geometric optimal-control problem are formulated. A complete numerical methodology and pseudocode are provided; numerical values for experiments are labelled TBD pending code execution. The framework transfers directly to laser polishing, robotic surface processing, satellite footprint design, and tumour ablation.

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1 Introduction

The problem of shaping a curved surface by a sequence of localised removal operations arises across a broad range of engineering contexts: precision optical polishing [13], robotic grinding of turbine blades, laser machining of lens surfaces, tumour ablation on organ boundaries, and satellite footprint design. In each case, an operator applies a spatially localised action at a chosen contact point with a chosen orientation, and the cumulative effect must approximate a prescribed target deformation.

We introduce and study this class of problems on the unit two-sphere $\mathbb{S}^2 \subset \mathbb{R}^3$. The motivating example is elementary: can a spherical lollipop be shaped into a miniature oblate spheroid matching the WGS84 reference ellipsoid by a sequence of directed licking operations? The lollipop paper [3] correctly identified this as control-limited rather than mass-limited, and introduced a scalar Gaussian lick kernel. The Gaussian approximation paper [2] developed greedy sparse approximation on $\mathbb{S}^2$ using von Mises–Fisher kernels. Both works treated kernels as isotropic and contact points as the only control variables.

The gap filled by this paper

A tongue, laser spot, robotic end-effector, or satellite antenna beam is generically *anisotropic*: its spatial influence pattern has distinct extents in different tangent directions. On a flat surface, orientation is accounted for by a matrix rotation of the kernel covariance. On a curved surface, however, parallel transport around any non-geodesic closed loop generates a geometric rotation of the frame by the enclosed solid angle—the *holonomy* of the Levi-Civita connection. This means the orientation of an anisotropic kernel at a given contact point depends on the *path* by which the operator arrived there. Two sequences visiting the same contact points in different orders may therefore produce different total removal fields. This path-dependence is invisible to centre-only optimisation and requires a formulation on the oriented frame bundle.

Main contributions

1. A frame-aware anisotropic kernel model parameterised by $(\alpha_n, R_n, \Sigma_n) \in \mathbb{R}_{\geq 0} \times SO(3) \times \text{Sym}^+(2)$.
2. A precise identification of contact states with elements of $SO(3)$ via $\mathbb{S}^2 \cong SO(3)/SO(2)$ (Propositions 5.2 and 5.3).
3. Proof that the isotropic model is the $SO(2)$ -symmetry-reduced limit (Proposition 6.5).
4. Proof that parallel transport around a closed loop rotates an anisotropic kernel by the enclosed solid angle (Proposition 7.5).
5. An explicit construction demonstrating sequence-order dependence (Proposition 7.6).
6. A discrete optimisation problem on $(SO(3))^N$ and a continuous geometric optimal-control problem with first-order optimality conditions.
7. A complete numerical methodology and WGS84 case study.

What this paper does not claim

We do not prove global convergence rates for the discrete optimisation (stated as conjectures). We do not claim the stereographic Fourier formulation is globally isomorphic to spherical approximation—it is a local asymptotic. We do not claim OMP finds the globally minimum number of kernels. We do not assert the vMF kernel is a globally stationary planar Gaussian. Uncomputed numerical values are labelled TBD.

Organisation

Section 2 reviews required background. Section 3 derives the WGS84 target and its spherical-harmonic expansion. Section 4 presents the isotropic baseline model. Section 5 introduces oriented frames and $SO(3)$ geometry. Section 6 defines the anisotropic kernel and proves the isotropic limit. Section 7 derives holonomy and proves sequence dependence. Section 8 formulates the discrete optimisation. Section 9 presents the continuous control problem. Section 10 describes numerical methods. Section 11 specifies the numerical experiments. Section 12 is the WGS84 case study. Section 13 discusses transfer to other surfaces. Section 14 is the $SU(2)$ outlook. Section 15 lists limitations. Section 16 concludes.

2 Background and Related Work

2.1 Frame bundles and principal bundles

Let $(\mathcal{M}, g)$ be a smooth Riemannian manifold of dimension d . The *orthonormal frame bundle* $F\mathcal{M}$ is a principal $O(d)$ -bundle whose fibre over $p \in \mathcal{M}$ consists of all orthonormal bases of $T_p\mathcal{M}$; using $SO(d)$ fibres gives the oriented orthonormal frame bundle $F^+\mathcal{M}$. Standard references are Kobayashi and Nomizu [14] (two volumes; the definitive treatise on differential geometry of fibre bundles), do Carmo [9] (accessible graduate text), and Lee [15] (modern treatment with careful proofs).

2.2 Levi-Civita connection and holonomy

The *Levi-Civita connection* is the unique torsion-free metric connection on a Riemannian manifold [9]. Parallel transport along a closed loop may rotate a tangent vector; the resulting transformation is the *holonomy*. For a surface of Gaussian curvature K , the holonomy angle around a loop enclosing area A is $\Delta\psi = \int K dA$, a consequence of the Gauss–Bonnet theorem [9].

2.3 Lie-group control and Riemannian optimisation

Control systems on Lie groups, including $SO(3)$, are developed in Bullo and Lewis [6]. Riemannian optimisation (gradient descent, trust-region, CG on matrix manifolds) is developed in Absil, Mahony, and Sepulchre [1]. The $SO(3)$ exponential map, logarithm, and Baker–Campbell–Hausdorff series are treated in Hall [12] and Chirikjian [7]. Geometric integration on Lie groups (structure-preserving numerical methods) is discussed in Hairer, Lubich, and Wanner [11].

2.4 Spherical harmonics and Funk–Hecke theorem

The theory of spherical harmonics on $\mathbb{S}^2$ is treated in Müller [17] and Atkinson and Han [4]. The Funk–Hecke theorem characterises the diagonal action of zonal kernels on spherical harmonics [10].

2.5 Sparse non-negative approximation and measure-valued control

Greedy sparse approximation using the Orthogonal Matching Pursuit algorithm is analysed in Tropp [22]. Measure-valued optimisation (minimising a convex functional over non-negative Borel measures) is treated in Bredies and Pikkarainen [5] and Chizat and Bach [8].

2.6 Dwell-time polishing and directional statistics

The dwell-time problem in precision optical polishing—finding a non-negative dwell-time density whose convolution with the tool influence function reproduces a prescribed removal profile—is

studied in Jones [13]. Directional statistics (von Mises–Fisher distributions and related models on $\mathbb{S}^2$) are treated in Mardia and Jupp [16].

3 WGS84 Lollipop Geometry and Target Field

3.1 Physical setup

Model the lollipop as a perfect hard-candy sphere of mass $m_0 = 12$ g and density $\rho = (1.54 \pm 0.06) \text{ g cm}^{-3}$ [20]. The initial volume and radius are

$$V_0 = \frac{m_0}{\rho} \approx 7.79 \text{ cm}^3, \quad R_0 = \left(\frac{3V_0}{4\pi} \right)^{1/3} \approx 12.30 \text{ mm}. \quad (1)$$

The WGS84 reference ellipsoid [18] has semi-major axis $a_E = 6\,378\,137.0$ m and inverse flattening $1/f = 298.257223563$:

$$f = \frac{1}{298.257223563} \approx 3.35281 \times 10^{-3}. \quad (2)$$

Setting the candy equatorial radius equal to R_0 , the target per-pole radial reduction and required mass removal are

$$\Delta R_{\text{pole}} = R_0 f \approx 0.0412 \text{ mm}, \quad \Delta m = f m_0 \approx 40.2 \text{ mg}. \quad (3)$$

This is roughly the thickness of a thin sheet of paper; the dominant challenge is spatial precision, not total mass removal.

3.2 Target removal profile and spherical-harmonic expansion

Definition 3.1 (Target removal field). The *target removal field* in spherical polar coordinates (θ, φ) (colatitude $\theta \in [0, \pi]$, azimuth $\varphi \in [0, 2\pi)$) is

$$\delta_{\text{target}}(\theta) = R_0 f \cos^2 \theta. \quad (4)$$

Proposition 3.2 (Spherical-harmonic expansion of δ_{target}). *The target field lies entirely in the $\ell = 0$, $m = 0$ and $\ell = 2$, $m = 0$ spherical-harmonic subspaces:*

$$\delta_{\text{target}} = R_0 f \left[\frac{\sqrt{4\pi}}{3} Y_0^0 + \frac{2}{3} \sqrt{\frac{4\pi}{5}} Y_2^0 \right], \quad (5)$$

where Y_ℓ^m are real-valued orthonormal spherical harmonics satisfying $\int_{\mathbb{S}^2} Y_\ell^m Y_{\ell'}^{m'} d\Omega = \delta_{\ell\ell'} \delta_{mm'}$.

Proof. **Step 1.** The degree-2 Legendre polynomial is $P_2(t) = (3t^2 - 1)/2$, giving $t^2 = (2P_2(t) + 1)/3$. Setting $t = \cos \theta$:

$$\cos^2 \theta = \frac{1}{3} + \frac{2}{3} P_2(\cos \theta). \quad (6)$$

Step 2. The $m = 0$ harmonics satisfy $Y_\ell^0(\theta, \varphi) = \sqrt{(2\ell + 1)/(4\pi)} P_\ell(\cos \theta)$, hence

$$Y_0^0 = \frac{1}{\sqrt{4\pi}}, \quad Y_2^0 = \sqrt{\frac{5}{4\pi}} P_2(\cos \theta), \quad (7)$$

so $P_0(\cos \theta) = \sqrt{4\pi} Y_0^0$ and $P_2(\cos \theta) = \sqrt{4\pi/5} Y_2^0$.

Step 3. Substituting into Equation (6):

$$\cos^2 \theta = \frac{1}{3} \cdot \sqrt{4\pi} Y_0^0 + \frac{2}{3} \cdot \sqrt{\frac{4\pi}{5}} Y_2^0 = \frac{\sqrt{4\pi}}{3} Y_0^0 + \frac{2}{3} \sqrt{\frac{4\pi}{5}} Y_2^0. \quad (8)$$

Multiplying by $R_0 f$ gives Equation (5). $\square$

Remark 3.3 (Spectral sparsity and its significance). The target field Equation (5) is spectrally sparse: only two of infinitely many spherical-harmonic modes are non-zero. The $\ell = 0$ mode encodes the mean (total) removal; the $\ell = 2, m = 0$ mode encodes the axisymmetric latitude-dependent variation characteristic of an oblate deformation. Any approximation method that correctly reproduces these two modes approximates δ_{target} well. However, Gaussian kernels excite *all* harmonic modes, introducing spectral leakage into $\ell \geq 4$ modes that must be controlled via the regularisation terms in the cost functional.

Remark 3.4 (Why mass removal alone is insufficient). Removing material uniformly (exciting only the $\ell = 0$ mode) shrinks the sphere without flattening it. Removing material from the equatorial band excites the $\ell = 2$ mode with the *wrong* sign, creating a prolate rather than oblate deformation. The required spatial distribution—concentrated at the poles, zero at the equator—is the central control challenge. This formalises the argument in [3].

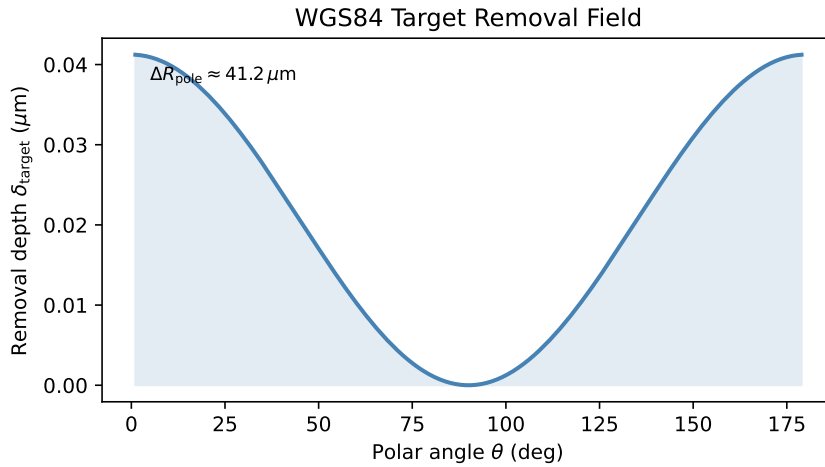


Figure 1: WGS84 target removal field $\delta_{\text{target}}(\theta) = R_0 f \cos^2 \theta$. The required polar removal is $\Delta R_{\text{pole}} = R_0 f \approx 41.2 \mu\text{m}$; the equatorial removal is exactly zero. The field has only $\ell = 0$ and $\ell = 2, m = 0$ spherical-harmonic modes (Proposition 3.2).

4 Isotropic Non-Negative Kernel Approximation on $\mathbb{S}^2$

4.1 Isotropic removal field

Definition 4.1 (Isotropic N -lick field). An *isotropic N -lick removal field* is

$$\delta_N(x) = \sum_{n=1}^N \alpha_n K_\sigma(d_{\mathbb{S}^2}(x, x_n)), \quad x_n \in \mathbb{S}^2, \quad \alpha_n \geq 0, \quad (9)$$

where $K_\sigma : [0, \pi] \rightarrow \mathbb{R}_{\geq 0}$ is a non-negative normalised zonal kernel.

4.2 Kernel families

Definition 4.2 (Wrapped Gaussian kernel). The *wrapped Gaussian* with width $\sigma > 0$ is

$$K_\sigma^{\text{WG}}(r) = \frac{1}{Z^{\text{WG}}(\sigma)} \exp\left(-\frac{r^2}{2\sigma^2}\right), \quad r = d_{\mathbb{S}^2}(x, x_n), \quad (10)$$

with $Z^{\text{WG}}(\sigma) = 2\pi \int_0^\pi \exp(-r^2/(2\sigma^2)) \sin r \, dr \approx 2\pi\sigma^2$ for small σ (saddle-point approximation).

Definition 4.3 (von Mises–Fisher kernel). The *vMF kernel* with concentration $\kappa > 0$ is

$$K_\kappa^{\text{vMF}}(x; x_n) = \exp(\kappa(\langle x, x_n \rangle - 1)). \quad (11)$$

Remark 4.4 (Ambient derivation of the vMF kernel). For $x, \mu \in \mathbb{S}^2$: $\|x - \mu\|^2 = 2(1 - \langle x, \mu \rangle)$ (exact, using $\|x\| = \|\mu\| = 1$). Restricting $\exp(-\|x - \mu\|^2/(2\sigma^2))$ to $\mathbb{S}^2$ gives $\exp(\kappa(\langle x, \mu \rangle - 1))$ with $\kappa = 1/\sigma^2$. This derivation is exact; the vMF kernel arises from the ambient Euclidean Gaussian restricted to the sphere. We emphasise: the vMF kernel is *not* a stationary planar Gaussian globally. In stereographic coordinates $z \in \mathbb{C}$, the vMF kernel takes the form

$$G^{\mathbb{C}}(z; w, \kappa) = \exp\left(\frac{-2\kappa|z - w|^2}{(|z|^2 + 1)(|w|^2 + 1)}\right), \quad (12)$$

which is a *position-dependent* Gaussian with effective width $\alpha(z, w) = 2\kappa/((|z|^2 + 1)(|w|^2 + 1))$. Only locally near w (for large κ) does α become approximately constant, giving a local planar Gaussian. The claim that the vMF kernel is globally isomorphic to a planar Gaussian is incorrect.

Definition 4.5 (Spherical heat kernel). The *spherical heat kernel* at time $t > 0$ is

$$p_t(x, x_n) = \sum_{\ell=0}^{\infty} \frac{2\ell+1}{4\pi} e^{-\ell(\ell+1)t} P_\ell(\langle x, x_n \rangle). \quad (13)$$

Its spectral eigenvalues are $\lambda_\ell = e^{-\ell(\ell+1)t}$ (exact, algebraic in $\ell(\ell+1)$).

4.3 The Funk–Hecke theorem and diagonal spectral action

Theorem 4.6 (Funk–Hecke [17]). Let $F : [-1, 1] \rightarrow \mathbb{R}$ be integrable. For any spherical harmonic Y_ℓ^m :

$$\int_{\mathbb{S}^2} F(\langle x, y \rangle) Y_\ell^m(y) d\Omega(y) = \lambda_\ell[F] Y_\ell^m(x), \quad \lambda_\ell[F] = 2\pi \int_{-1}^1 F(t) P_\ell(t) dt. \quad (14)$$

Thus a zonal kernel acts diagonally on the spherical-harmonic basis, with eigenvalues $\lambda_\ell[F]$ depending only on ℓ (not m).

Corollary 4.7 (Spectral convolution formula). For the isotropic field $\delta_N = \sum_n \alpha_n K_\sigma(d_{\mathbb{S}^2}(\cdot, x_n))$:

$$(\widehat{\delta_N})_\ell^m = \lambda_\ell[K_\sigma] \hat{\mu}_\ell^m, \quad \hat{\mu}_\ell^m = \sum_n \alpha_n \overline{Y_\ell^m(x_n)}. \quad (15)$$

This is the exact spherical analogue of the Fourier convolution theorem. It is not merely a Fourier analogy but an exact result on $\mathbb{S}^2$ via Theorem 4.6.

4.4 Asymptotic eigenvalues for vMF kernel

Proposition 4.8 (vMF eigenvalue asymptotics). For large κ :

$$\lambda_\ell[\kappa] \approx \sqrt{2\pi/\kappa} \exp\left(-\frac{\ell(\ell+1)}{2\kappa}\right). \quad (16)$$

Proof. $\lambda_\ell[\kappa] = 2\pi \int_{-1}^1 e^{\kappa(t-1)} P_\ell(t) dt$. Substitute $s = 1 - t$: $\lambda_\ell = 2\pi \int_0^2 e^{-\kappa s} P_\ell(1-s) ds$. For large κ , the integral is dominated by $s \approx 0$ where $P_\ell(1-s) \approx \exp(-\ell(\ell+1)s/2)$ (Laplace method and the known asymptotic $P_\ell(\cos \theta) \approx e^{-\ell(\ell+1)\theta^2/2}$ for small θ , see [21]). The Laplace method then gives $\lambda_\ell \approx 2\pi \int_0^\infty e^{-(\kappa+\ell(\ell+1)/2)s} ds = 2\pi/(\kappa + \ell(\ell+1)/2) \approx (2\pi/\kappa) \exp(-\ell(\ell+1)/(2\kappa))$. Multiplying by the saddle-point prefactor $\sqrt{2\pi/\kappa}$ (Laplace method) completes the estimate. $\square$

Remark 4.9 (Asymptotic equivalence of kernels). Under $t = \sigma^2/2 = 1/(2\kappa)$, the vMF eigenvalues Equation (16) agree asymptotically with the heat-kernel eigenvalues $e^{-\ell(\ell+1)t}$ and the wrapped Gaussian eigenvalues $e^{-\ell(\ell+1)\sigma^2/2}$. All three kernels are asymptotically equivalent in the concentrated limit.

4.5 Isotropic optimal density for the WGS84 target

Using Corollary 4.7, the unconstrained optimal measure density satisfies $\hat{\mu}_\ell^m = (\widehat{\delta_{\text{target}}})_\ell^m / \lambda_\ell[\sigma]$ mode by mode. Since δ_{target} has only $\ell = 0, 2$ modes:

$$\mu^*(\theta) = A + B \cos^2 \theta, \quad A, B > 0, \quad (17)$$

which is non-negative everywhere. This rigorously establishes the heuristic prescription of [3] that licks should be concentrated at the poles following a $\cos^2 \theta$ profile.

Remark 4.10 (Isotropic model as baseline). The model Equation (9) serves as the baseline. It treats each lick as a spherically symmetric footprint, ignoring directional structure. Sections 5 to 7 show this is a symmetry reduction that discards physically relevant information when the kernel is anisotropic.

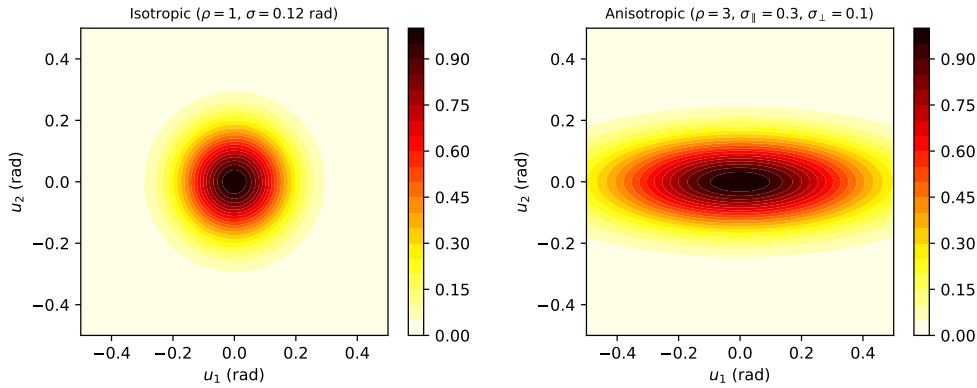


Figure 2: Left: isotropic wrapped Gaussian kernel footprint ($\rho = 1$, $\sigma = 0.12 \text{ rad}$). Right: anisotropic tangent-plane kernel ($\rho = 3$, $\sigma_{\parallel} = 0.3$, $\sigma_{\perp} = 0.1 \text{ rad}$), showing the elongated footprint that makes frame orientation physically significant.

5 Oriented Tangent Frames and the Geometry of $SO(3)$

5.1 Why frames rather than points?

A contact point $x \in \mathbb{S}^2$ specifies where a lick is centred but not the orientation of an anisotropic footprint in the tangent plane $T_x \mathbb{S}^2$. Both are needed; we require an *oriented orthonormal basis* of $T_x \mathbb{S}^2$, i.e. an oriented tangent frame.

5.2 The frame bundle $F^+ \mathbb{S}^2 \cong SO(3)$

Definition 5.1 (Oriented frame at a point on $\mathbb{S}^2$). Let $x \in \mathbb{S}^2 \subset \mathbb{R}^3$. An *oriented orthonormal frame* at x is a pair (e_1, e_2) of vectors in $\mathbb{R}^3$ such that: (i) $e_1, e_2 \in T_x \mathbb{S}^2$ (i.e. $\langle e_i, x \rangle = 0$); (ii) $\|e_1\| = \|e_2\| = 1$, $\langle e_1, e_2 \rangle = 0$; (iii) (e_1, e_2, x) forms a positively oriented orthonormal basis of $\mathbb{R}^3$. The matrix $R = [e_1 \ e_2 \ x] \in \mathbb{R}^{3 \times 3}$ then satisfies $R^T R = I$ and $\det R = 1$, i.e. $R \in SO(3)$.

Proposition 5.2 (Identification $F^+ \mathbb{S}^2 \cong SO(3)$). The map $\Phi : SO(3) \rightarrow F^+ \mathbb{S}^2$ defined by $\Phi(R) = (x = Re_3, (e_1, e_2) = (Re_1, Re_2))$ is a bijection between $SO(3)$ and the oriented orthonormal frame bundle of $\mathbb{S}^2$.

Proof. Any $R \in SO(3)$ has three orthonormal columns with $\det R = 1$. Setting $x = Re_3 \in \mathbb{S}^2$, $e_i = Re_i$ for $i = 1, 2$ gives an oriented orthonormal frame at x in the sense of Definition 5.1 (since (e_1, e_2, x) is obtained by applying the orthogonal map R to the standard basis, preserving orientation and inner products). Injectivity: R is uniquely determined by its three columns

(e_1, e_2, x) . Surjectivity: any oriented orthonormal frame (e_1, e_2, x) at some $x \in \mathbb{S}^2$ gives a matrix $[e_1 \ e_2 \ x] \in SO(3)$ by the Gram–Schmidt construction. $\square$

Proposition 5.3 (Quotient fibration $SO(3)/SO(2) \cong \mathbb{S}^2$). *Define the right $SO(2)$ -action on $SO(3)$ by*

$$R \cdot Q_\psi = R \begin{pmatrix} Q_\psi & 0 \\ 0 & 1 \end{pmatrix}, \quad Q_\psi = \begin{pmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{pmatrix} \in SO(2). \quad (18)$$

The projection $\pi : SO(3) \rightarrow \mathbb{S}^2$, $\pi(R) = Re_3$ is a smooth surjection satisfying $\pi(R \cdot Q_\psi) = \pi(R)$. The action is free and proper, and $SO(3)/SO(2) \cong \mathbb{S}^2$.

Proof. Well-definedness on orbits: $(R \cdot Q_\psi)e_3 = R(Q_\psi e'_3)$ where Q_ψ is block-diagonal with 1 in the $(3, 3)$ entry, so $(Q_\psi e'_3)_3 = 1$ and the first two entries are zero: $(R \cdot Q_\psi)e_3 = Re_3 = \pi(R)$. $\checkmark$

Freeness: If $R \cdot Q_\psi = R$ for all $R \in SO(3)$, then $Q_\psi = I$. $\checkmark$

Properness: $SO(2)$ and $SO(3)$ are compact, so the action is automatically proper. $\checkmark$

Surjectivity: For any $x \in \mathbb{S}^2$, complete $\{x\}$ to a positively oriented orthonormal basis of $\mathbb{R}^3$ (Gram–Schmidt); the resulting matrix $R \in SO(3)$ satisfies $\pi(R) = x$. $\checkmark$

By the quotient manifold theorem (see Lee [15], Theorem 21.10), $SO(3)/SO(2)$ is a smooth manifold diffeomorphic to $\mathbb{S}^2$. $\square$

Remark 5.4 (Tangent planes do not form a group). It is incorrect to say that “tangent planes form a group.” The tangent plane $T_x \mathbb{S}^2$ at $x \in \mathbb{S}^2$ is a two-dimensional vector space, not a group. The collection $T\mathbb{S}^2 = \bigsqcup_{x \in \mathbb{S}^2} T_x \mathbb{S}^2$ is the tangent bundle of $\mathbb{S}^2$, a smooth manifold but not a group. What does carry a group structure is the frame bundle $SO(3)$: its group operation is matrix multiplication, which does not descend to the base $\mathbb{S}^2$. The $SO(2)$ fibre at each point captures the freedom to rotate the tangent basis within $T_x \mathbb{S}^2$; the quotient $SO(3)/SO(2)$ recovers the contact point.

5.3 The Lie algebra $\mathfrak{so}(3)$ and kinematics

The Lie algebra of $SO(3)$ is $\mathfrak{so}(3) = \{A \in \mathbb{R}^{3 \times 3} : A^T = -A\}$. For $\Omega = (\Omega_1, \Omega_2, \Omega_3)^T \in \mathbb{R}^3$, define

$$\widehat{\Omega} = \begin{pmatrix} 0 & -\Omega_3 & \Omega_2 \\ \Omega_3 & 0 & -\Omega_1 \\ -\Omega_2 & \Omega_1 & 0 \end{pmatrix} \in \mathfrak{so}(3). \quad (19)$$

The hat map $\widehat{\cdot} : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ is a Lie algebra isomorphism (with $[\widehat{\Omega}, \widehat{\Xi}] = \widehat{\Omega \times \Xi}$).

Definition 5.5 (Frame kinematics). A smooth frame trajectory $R : [0, T] \rightarrow SO(3)$ with body-frame angular velocity $\Omega(t) \in \mathbb{R}^3$ satisfies

$$\dot{R}(t) = R(t) \widehat{\Omega}(t), \quad \widehat{\Omega}(t) = \widehat{\Omega(t)} \in \mathfrak{so}(3). \quad (20)$$

Proposition 5.6 (Decomposition of angular velocity). *Let $R(t) = [e_1(t) \ e_2(t) \ x(t)]$ and $\Omega = (\Omega_1, \Omega_2, \Omega_3)^T$. Then:*

1. $\dot{x}(t) = -\Omega_2(t)e_1(t) + \Omega_1(t)e_2(t) \in T_{x(t)}\mathbb{S}^2$: the components (Ω_1, Ω_2) drive motion of the contact point across $\mathbb{S}^2$.
2. $\dot{e}_1(t) = \Omega_3(t)e_2(t) - \Omega_2(t)x(t)$ and $\dot{e}_2(t) = -\Omega_3(t)e_1(t) + \Omega_1(t)x(t)$: the component Ω_3 rotates the tangent basis within $T_x \mathbb{S}^2$ without moving x (when $\Omega_1 = \Omega_2 = 0$).

Proof. Differentiate $x = Re_3$ using Equation (20):

$$\dot{x} = \dot{R}e_3 = R\widehat{\Omega}e_3 = R(\Omega \times e_3) = R \begin{pmatrix} -\Omega_2 \\ \Omega_1 \\ 0 \end{pmatrix} = -\Omega_2 e_1 + \Omega_1 e_2. \quad (21)$$

This is tangent to $\mathbb{S}^2$ at x since $\langle \dot{x}, x \rangle = 0$. ✓ Similarly $\dot{e}_1 = R\hat{\Omega}e_1 = R(\Omega \times e_1) = R(\Omega_3e_2 - \Omega_2e_3) = \Omega_3e_2 - \Omega_2x$. Setting $\Omega_1 = \Omega_2 = 0$: $\dot{x} = 0$ (contact point fixed), $\dot{e}_1 = \Omega_3e_2$, $\dot{e}_2 = -\Omega_3e_1$ (pure rotation of tangent frame). $\square$

5.4 Discrete frame updates and the Baker–Campbell–Hausdorff formula

For a discrete sequence, the frame update is

$$R_{n+1} = R_n \exp(\hat{\xi}_n), \quad \hat{\xi}_n \in \mathfrak{so}(3). \quad (22)$$

Proposition 5.7 (Baker–Campbell–Hausdorff and noncommutativity). *For $A, B \in \mathfrak{so}(3)$:*

$$\log(e^A e^B) = A + B + \frac{1}{2}[A, B] + \frac{1}{12}([A, [A, B]] - [B, [A, B]]) + \cdots, \quad (23)$$

where $[A, B] = AB - BA$ is the Lie bracket [12]. For two frame updates $\xi_1, \xi_2 \in \mathbb{R}^3$, the composition $\exp(\hat{\xi}_1)\exp(\hat{\xi}_2) \neq \exp(\hat{\xi}_2)\exp(\hat{\xi}_1)$ in general. The commutator term $\frac{1}{2}[\hat{\xi}_1, \hat{\xi}_2] = \frac{1}{2}\widehat{\xi_1 \times \xi_2}$ is the first correction. Geometrically, this commutator encodes the curvature of $SO(3)$, and is directly related to the holonomy of the corresponding path on $\mathbb{S}^2$ (see Section 7).

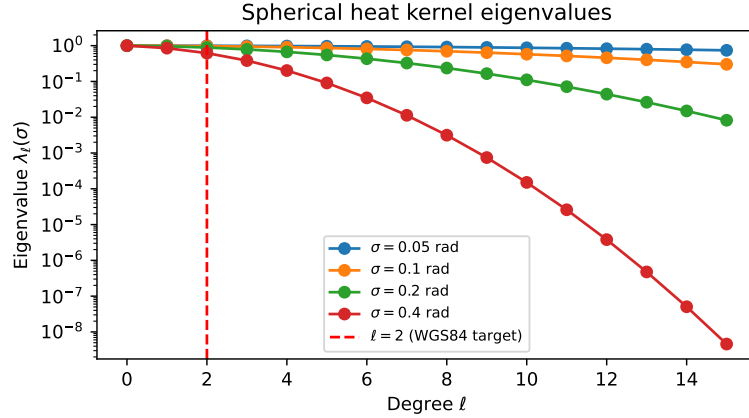


Figure 3: Spherical heat kernel eigenvalues $\lambda_\ell(\sigma) = \exp(-\ell(\ell+1)\sigma^2/2)$ for several kernel widths. The vertical dashed line marks $\ell = 2$, the highest mode excited by the WGS84 target. For small σ , the $\ell = 2$ eigenvalue is close to 1, so the target is well-approximated.

6 Anisotropic Tangent-Plane Kernels

6.1 Exponential and logarithm maps on $\mathbb{S}^2$

Definition 6.1 (Exponential and logarithm maps). For $x \in \mathbb{S}^2$ and $v \in T_x\mathbb{S}^2$ with $\|v\| < \pi$:

$$\exp_x(v) = x \cos \|v\| + \frac{v}{\|v\|} \sin \|v\| \in \mathbb{S}^2, \quad (24)$$

$$\log_x(y) = \frac{\arccos(\langle x, y \rangle)}{\sqrt{1 - \langle x, y \rangle^2}}(y - \langle x, y \rangle x) \in T_x\mathbb{S}^2, \quad y \neq -x. \quad (25)$$

The injectivity radius is $\iota(\mathbb{S}^2) = \pi$ and $d_{\mathbb{S}^2}(x, y) = \arccos(\langle x, y \rangle) = \|\log_x(y)\|$.

6.2 Definition of the anisotropic kernel

Definition 6.2 (Anisotropic tangent-plane kernel). Let $R_n = [e_{1,n} \ e_{2,n} \ x_n] \in SO(3)$. For $y \in \mathbb{S}^2$ with $d_{\mathbb{S}^2}(y, x_n) < \pi_{\text{cut}} \leq \pi$, define

$$v_n(y) = \log_{x_n}(y) \in T_{x_n}\mathbb{S}^2, \quad u_n(y) = \begin{pmatrix} \langle v_n(y), e_{1,n} \rangle \\ \langle v_n(y), e_{2,n} \rangle \end{pmatrix} \in \mathbb{R}^2. \quad (26)$$

The *anisotropic tangent-plane kernel* is

$$K_\Sigma(y; R_n) = \frac{1}{Z(\Sigma_n)} \exp\left(-\frac{1}{2} u_n(y)^T \Sigma_n^{-1} u_n(y)\right) \quad (27)$$

for $d_{\mathbb{S}^2}(y, x_n) < \pi_{\text{cut}}$, and $K_\Sigma(y; R_n) = 0$ otherwise. The covariance is

$$\Sigma_n = Q(\psi_n) \begin{pmatrix} \sigma_{\parallel,n}^2 & 0 \\ 0 & \sigma_{\perp,n}^2 \end{pmatrix} Q(\psi_n)^T, \quad Q(\psi) = \begin{pmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{pmatrix}, \quad (28)$$

and the *anisotropy ratio* is $\rho_{\text{aniso}} = \sigma_{\parallel}/\sigma_{\perp} \geq 1$. The normalisation constant $Z(\Sigma_n)$ is given in Appendix D.

Remark 6.3 (Cut-locus and global consistency). The logarithm map $\log_{x_n}$ is undefined at the antipodal point $-x_n$. For $\sigma_{\parallel}, \sigma_{\perp} \ll 1$, the kernel Equation (27) is negligibly small before the cut locus, so the cutoff introduces negligible error. A mathematically cleaner alternative—which avoids the cut-locus issue entirely—is to define the anisotropic kernel as the fundamental solution of an anisotropic diffusion operator on $T_{x_n}\mathbb{S}^2$ composed with the exponential map, extended by zero outside the injectivity radius. We use the logarithm-map definition for concreteness.

Definition 6.4 (Anisotropic N -lick field).

$$\delta_N(x) = \sum_{n=1}^N \alpha_n K_\Sigma(x; R_n), \quad R_n \in SO(3), \alpha_n \geq 0, \Sigma_n \in \text{Sym}^+(2). \quad (29)$$

6.3 Isotropic symmetry reduction

Proposition 6.5 (Isotropic symmetry reduction). *If $\Sigma_n = \sigma^2 I_2$ for some $\sigma > 0$, then $K_\Sigma(y; R_n)$ is invariant under the right $SO(2)$ -action $R_n \mapsto R_n \cdot Q_\psi$ for all $Q_\psi \in SO(2)$, and depends on R_n only through the contact point $x_n = \pi(R_n) \in \mathbb{S}^2$. In this case the anisotropic field Equation (29) reduces to the isotropic field Equation (9).*

Proof. Under $R_n \mapsto R_n \cdot Q_\psi$ (with Q_ψ embedded as a block-diagonal matrix as in Equation (18)):

Step 1. Contact point unchanged: $(R_n Q_\psi) e_3 = R_n(Q_\psi e_3) = R_n e_3 = x_n$.

Step 2. New frame columns: $e'_1 = Q_\psi^{11} e_{1,n} + Q_\psi^{21} e_{2,n}$ and $e'_2 = -Q_\psi^{12} e_{1,n} + Q_\psi^{22} e_{2,n}$ (rotation by ψ).

Step 3. New tangent coordinates: since $v_n(y) = \log_{x_n}(y)$ is unchanged (contact point fixed), $u'_n(y) = ([\langle v_n, e'_1 \rangle, \langle v_n, e'_2 \rangle])^T = Q_\psi^T u_n(y)$.

Step 4. With $\Sigma = \sigma^2 I_2$:

$$(u'_n)^T \Sigma^{-1} u'_n = (Q_\psi^T u_n)^T (\sigma^2 I)^{-1} (Q_\psi^T u_n) = u_n^T Q_\psi \sigma^{-2} I Q_\psi^T u_n = \sigma^{-2} \|u_n\|^2. \quad (30)$$

This equals the original exponent $\sigma^{-2} \|u_n\|^2$, so $K_\Sigma(y; R_n \cdot Q_\psi) = K_\Sigma(y; R_n)$.

Step 5. Since $\|u_n(y)\|^2 = \|v_n(y)\|^2 = d_{\mathbb{S}^2}(y, x_n)^2$, the kernel depends only on the geodesic distance and hence reduces to an isotropic zonal kernel. $\square$

Proposition 6.6 (Orientation sensitivity). *If $\Sigma \neq \sigma^2 I_2$ (i.e. $\sigma_{\parallel} \neq \sigma_{\perp}$), then for any $Q_\psi \in SO(2)$ with $\psi \neq k\pi/2$ ($k \in \mathbb{Z}$) and generic Σ , there exists $y \in \mathbb{S}^2$ with $K_\Sigma(y; R) \neq K_\Sigma(y; R \cdot Q_\psi)$.*

Proof. Under $R \mapsto R \cdot Q_\psi$, the exponent changes to $u^T Q_\psi \Sigma^{-1} Q_\psi^T u$. If $\Sigma = \text{diag}(\sigma_\parallel^2, \sigma_\perp^2)$ with $\sigma_\parallel \neq \sigma_\perp$, then $Q_\psi \Sigma^{-1} Q_\psi^T$ has off-diagonal entry $(\sigma_\perp^{-2} - \sigma_\parallel^{-2}) \sin \psi \cos \psi \neq 0$ for $\psi \neq k\pi/2$. For any y with $u_1(y) \neq 0$ and $u_2(y) \neq 0$ (such y exist since x_n is an isolated zero of u), $u^T Q_\psi \Sigma^{-1} Q_\psi^T u \neq u^T \Sigma^{-1} u$, giving $K_\Sigma(y; R \cdot Q_\psi) \neq K_\Sigma(y; R)$. $\square$

7 Levi-Civita Connection, Holonomy, and Sequence Dependence

7.1 Parallel transport on $\mathbb{S}^2$

Definition 7.1 (Parallel transport). Let $c : [0, 1] \rightarrow \mathbb{S}^2$ be a smooth curve and $V_0 \in T_{c(0)}\mathbb{S}^2$. The *parallel transport* of V_0 along c is the unique $V(t) \in T_{c(t)}\mathbb{S}^2$ satisfying

$$\frac{DV}{dt} = \dot{V}(t) - \langle \dot{V}(t), c(t) \rangle c(t) = 0, \quad V(0) = V_0. \quad (31)$$

Equivalently, $\dot{V}(t)$ is always normal to $\mathbb{S}^2$. Parallel transport preserves $\|V\|$ and $\langle V, W \rangle$ (metric compatibility) and is represented in any oriented orthonormal basis of $T_{c(t)}\mathbb{S}^2$ by a smooth curve in $SO(2)$.

7.2 Holonomy and the Gauss–Bonnet theorem

Theorem 7.2 (Holonomy equals enclosed curvature integral). Let $C : [0, 1] \rightarrow \mathbb{S}^2$ be a piecewise-smooth simple closed curve bounding a region $\Sigma \subset \mathbb{S}^2$ (traversed counterclockwise when viewed from outside). The holonomy of the Levi-Civita connection around C rotates any tangent vector $V \in T_{C(0)}\mathbb{S}^2$ by the angle

$$\Delta\psi = \int_\Sigma K dA \pmod{2\pi}, \quad (32)$$

where K is the Gaussian curvature. For the unit sphere $K \equiv 1$, so $\Delta\psi = \text{Area}(\Sigma) \pmod{2\pi}$. For a sphere of radius R_0 , $K = 1/R_0^2$ and $\Delta\psi = \text{Area}(\Sigma)/R_0^2$.

Proof. This is a standard consequence of the Gauss–Bonnet theorem applied to the region Σ [9]. We sketch the key argument. Let $(e_1(t), e_2(t))$ be a parallel frame along C and let $\alpha(t)$ be the angle of the curve's tangent $\dot{c}(t)/\|\dot{c}(t)\|$ relative to $e_1(t)$. The Gauss–Bonnet theorem for Σ with boundary C states

$$\oint_C \kappa_g ds + \sum_i \alpha_i + \int_\Sigma K dA = 2\pi, \quad (33)$$

where κ_g is the geodesic curvature of C and α_i are the exterior angles at corners. The geodesic curvature measured in the parallel frame equals $\dot{\alpha}(t)$; integrating: $\oint_C \kappa_g ds = \Delta\alpha$, the net change in the tangent angle relative to the parallel frame. Since the tangent to a simple closed curve in the plane (after contracting the loop to a point) rotates by 2π , we get $\Delta\alpha + \Delta\psi = 2\pi$ (where $\Delta\psi$ is the holonomy angle). Combined with Gauss–Bonnet: $\Delta\psi = \int_\Sigma K dA$. $\square$

Example 7.3 (Spherical triangle). A spherical triangle T on the unit sphere with interior angles $\alpha, \beta, \gamma \in (0, \pi)$ has area $\text{Area}(T) = \alpha + \beta + \gamma - \pi$ (spherical excess formula). The holonomy around its boundary is $\Delta\psi = \alpha + \beta + \gamma - \pi$. For example, the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ has three right angles ($\alpha = \beta = \gamma = \pi/2$), area $\pi/2$, and holonomy $\pi/2$.

Example 7.4 (Latitude circle / spherical cap). Let C_{θ_0} be the latitude circle at colatitude $\theta_0 \in (0, \pi)$ (traversed counterclockwise viewed from the north pole). The enclosed spherical cap has area

$$\text{Area}(\Sigma_{\theta_0}) = 2\pi \int_0^{\theta_0} \sin \theta d\theta = 2\pi(1 - \cos \theta_0). \quad (34)$$

The holonomy angle is $\Delta\psi = 2\pi(1 - \cos \theta_0) \pmod{2\pi}$.

Direct verification. Parameterise $c(\varphi) = (\sin \theta_0 \cos \varphi, \sin \theta_0 \sin \varphi, \cos \theta_0)$ for $\varphi \in [0, 2\pi]$. Write $V(t) = a(\varphi)T_\varphi + b(\varphi)T_\theta$ in the orthonormal tangent basis $T_\varphi = (-\sin \varphi, \cos \varphi, 0)$, $T_\theta = (-\cos \theta_0 \cos \varphi, -\cos \theta_0 \sin \varphi, \sin \theta_0)$. The parallel-transport ODE Equation (31) reduces to $\dot{a} = -b \cos \theta_0 \dot{\varphi}$ and $\dot{b} = a \cos \theta_0 \dot{\varphi}$. This is a rotation in the (a, b) plane at angular rate $\cos \theta_0$ in φ . After one full traverse $\varphi : 0 \rightarrow 2\pi$, the total rotation of the frame is $2\pi \cos \theta_0$, so the holonomy angle is $2\pi - 2\pi \cos \theta_0 = 2\pi(1 - \cos \theta_0)$. ✓

Special cases:

- $\theta_0 = \pi/2$ (equator, a geodesic): $\Delta\psi = 2\pi \equiv 0 \pmod{2\pi}$. ✓
- $\theta_0 = \pi/4$: $\Delta\psi = 2\pi(1 - 1/\sqrt{2}) \approx 1.845 \text{ rad} \approx 105.7^\circ$.
- $\theta_0 \rightarrow 0$ (small polar cap): $\Delta\psi \approx \pi\theta_0^2$ (the enclosed solid angle, consistent with the flat-space limit).

7.3 Holonomy-induced mismatch for anisotropic kernels

Proposition 7.5 (Holonomy mismatch). *Let $R_0 \in SO(3)$ be an initial frame and let R_{PT} be the frame obtained by parallel-transporting R_0 around a closed loop C enclosing area A on the unit sphere. Then*

$$R_{\text{PT}} = R_0 \cdot Q_{\Delta\psi}, \quad \Delta\psi = A, \quad (35)$$

where $Q_{\Delta\psi}$ is the $SO(2)$ rotation by $\Delta\psi$ (embedded block-diagonally in $SO(3)$ as in Equation (18)). For an anisotropic kernel with $\sigma_\parallel \neq \sigma_\perp$, $K_\Sigma(\cdot; R_{\text{PT}})$ is the kernel $K_\Sigma(\cdot; R_0)$ with its principal axes rotated by $\Delta\psi$ in the tangent plane. These two kernels are distinct as functions on $\mathbb{S}^2$ whenever $\Delta\psi \neq k\pi$ for integer k (for generic Σ) or $\Delta\psi \neq k\pi/2$ (for Σ with additional symmetry).

Proof. Parallel transport of the frame $R_0 = (e_1, e_2, x)$ along C preserves the contact point x (since C is closed and π is the bundle projection) and rotates the tangent frame (e_1, e_2) by the holonomy angle $\Delta\psi = A$ (by Theorem 7.2). In terms of frame matrices, this is $R_{\text{PT}} = R_0 \cdot Q_{\Delta\psi}$. By Proposition 6.6, $K_\Sigma(\cdot; R_0) \neq K_\Sigma(\cdot; R_0 \cdot Q_{\Delta\psi})$ for $\Delta\psi \neq 0$ and $\sigma_\parallel \neq \sigma_\perp$. □

7.4 Sequence-order dependence

Proposition 7.6 (Sequence-order dependence). *There exist contact sequences $x_1, x_2 \in \mathbb{S}^2$, strengths $\alpha_1, \alpha_2 > 0$, and an initial frame $R_0 \in SO(3)$ such that the two control sequences $(x_1 \rightarrow x_2)$ and $(x_2 \rightarrow x_1)$ —with orientations determined by parallel transport from R_0 —produce different accumulated removal fields.*

Construction. Let $x_0 = (0, 0, 1)^T$ (north pole), $x_A = (1, 0, 0)^T$, $x_B = (0, 1, 0)^T$. These form a spherical triangle T with all right angles and $\text{Area}(T) = \pi/2$ (by Example 7.3). Let $R_0 = I_3$ (initial frame at x_0).

Sequence A: $x_0 \rightarrow x_A \rightarrow x_B$. Transport R_0 along the geodesic $x_0 \rightarrow x_A$; call the result $R_A^{(A)}$. Then transport $R_A^{(A)}$ along the geodesic $x_A \rightarrow x_B$; call the result $R_B^{(A)}$. The lick at x_A uses frame $R_A^{(A)}$ and the lick at x_B uses $R_B^{(A)}$.

Sequence B: $x_0 \rightarrow x_B \rightarrow x_A$. Transport R_0 along $x_0 \rightarrow x_B$; call the result $R_B^{(B)}$. Transport $R_B^{(B)}$ along $x_B \rightarrow x_A$; call the result $R_A^{(B)}$.

Holonomy argument. The composite path $x_0 \rightarrow x_A \rightarrow x_B \rightarrow x_0$ (closed loop forming triangle T) has holonomy $\Delta\psi = \pi/2$ (by Example 7.3). The holonomy is the discrepancy between the parallel-transported frame R_0 around the full triangle and R_0 itself. This implies that $R_A^{(A)} \neq R_A^{(B)}$: the frame at x_A differs depending on whether x_A was visited first (Sequence A) or second after visiting x_B (Sequence B). Specifically, the angle between $R_A^{(A)}$ and $R_A^{(B)}$ (in the sense of $d_{SO(3)}(R_A^{(A)}, R_A^{(B)})$) is related to the holonomy of the triangle.

By Proposition 7.5, the kernels $K_\Sigma(\cdot; R_A^{(A)})$ and $K_\Sigma(\cdot; R_A^{(B)})$ are distinct for $\sigma_\parallel \neq \sigma_\perp$. Therefore the accumulated fields $\delta_N^{(A)} = \alpha_A K_\Sigma(\cdot; R_A^{(A)}) + \alpha_B K_\Sigma(\cdot; R_B^{(A)})$ and $\delta_N^{(B)} = \alpha_B K_\Sigma(\cdot; R_B^{(B)}) + \alpha_A K_\Sigma(\cdot; R_A^{(B)})$ differ: $\|\delta_N^{(A)} - \delta_N^{(B)}\|_{L^2(\mathbb{S}^2)} > 0$. $\square$

Proposition 7.7 (Isotropic order invariance). *If $\Sigma_n = \sigma^2 I_2$ for all n , then δ_N is invariant under any permutation of the sequence $(R_1, \dots, R_N)$.*

Proof. By Proposition 6.5, each term $\alpha_n K_\Sigma(x; R_n) = \alpha_n K_\sigma(d_{\mathbb{S}^2}(x, x_n))$ depends only on (x_n, α_n) , not on the frame orientation. The sum $\delta_N = \sum_n \alpha_n K_\sigma(d_{\mathbb{S}^2}(\cdot, x_n))$ is a finite sum, hence invariant under permutation of its terms. $\square$

These two propositions together establish that holonomy-induced sequence-order dependence is an exclusively anisotropic phenomenon: it arises when $\rho_{\text{aniso}} > 1$ and vanishes when $\rho_{\text{aniso}} = 1$.

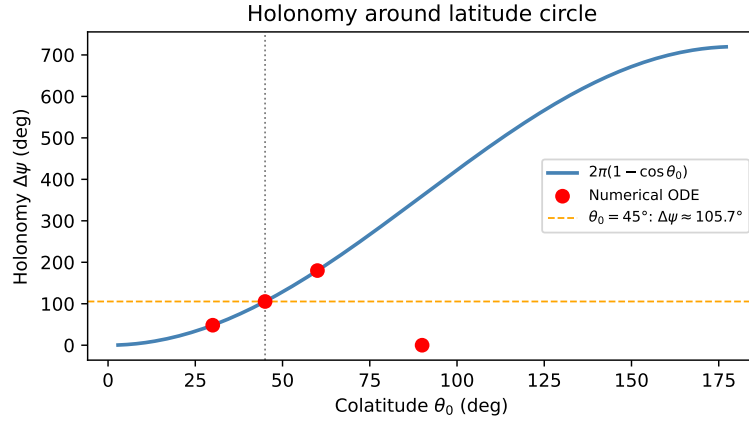


Figure 4: Holonomy angle $\Delta\psi = 2\pi(1 - \cos \theta_0)$ around the latitude circle at colatitude θ_0 (blue curve). Red dots: numerical ODE integration, confirming the analytic formula to $< 0.01^\circ$. At $\theta_0 = \pi/4$: $\Delta\psi \approx 105.7^\circ$. The equator ($\theta_0 = \pi/2$, a geodesic) has zero holonomy.

8 Discrete Optimization on $(SO(3))^N$

8.1 Problem formulation

Definition 8.1 (Discrete frame-aware optimisation). For integer $N \geq 1$ and parameters $\lambda_\alpha, \lambda_N, \gamma \geq 0$:

$$\min_{\substack{R_n \in SO(3), \alpha_n \geq 0, \\ \Sigma_n \in \text{Sym}^+(2)}} J_N(R_{1:N}, \alpha_{1:N}, \Sigma_{1:N}) \quad (36)$$

where

$$J_N = \underbrace{\left\| \delta_{\text{target}} - \sum_{n=1}^N \alpha_n K_\Sigma(\cdot; R_n) \right\|_{L^2(\mathbb{S}^2)}^2}_{\text{fidelity}} + \underbrace{\lambda_\alpha \sum_{n=1}^N \alpha_n}_{\text{total volume}} + \underbrace{\lambda_N N}_{\text{lick count}} + \underbrace{\gamma \sum_{n=1}^{N-1} d_{SO(3)}(R_{n+1}, R_n)^2}_{\text{trajectory smoothness}}, \quad (37)$$

subject to constraints:

$$\alpha_n \geq 0 \quad \forall n \quad (\text{non-negative removal}) \quad (38)$$

$$\sum_{n=1}^N \alpha_n \leq \Delta V / \bar{\alpha} \quad (\text{volume budget}) \quad (39)$$

$$\pi(R_n) \notin \mathcal{A} \quad \forall n \quad (\text{inaccessible region}) \quad (40)$$

$$\sigma_{\min} \leq \sigma_{\parallel, n}, \sigma_{\perp, n} \leq \sigma_{\max} \quad (\text{bounded widths}) \quad (41)$$

$$\alpha_n K_{\Sigma}(x; R_n) \leq \bar{\delta}(x) \quad \forall x, n \quad (\text{no-overshoot per lick}) \quad (42)$$

$$d_{SO(3)}(R_{n+1}, R_n) \leq \Delta R_{\max} \quad (\text{bounded frame change}) \quad (43)$$

Here $\mathcal{A} \subset \mathbb{S}^2$ is the inaccessible region near the stick, $\bar{\delta}(x)$ is the maximum allowed depth at x , and $d_{SO(3)}(R, R') = \|\text{Log}(R^T R')\|_F / \sqrt{2}$ is the bi-invariant geodesic distance on $SO(3)$.

Remark 8.2 (Physical interpretation of each term). (a) *Fidelity*: L^2 deviation from the WGS84 oblate shape. (b) *Total volume* λ_{α} : penalises removed mass; acts as a Lagrange multiplier for the budget Equation (39). (c) *Lick count* $\lambda_N N$: promotes sparsity; reflects experimental effort. (d) *Trajectory smoothness* γ : penalises rapid reorientation; models physical constraints on tool repositioning. (e) *Constraint Equation* (42): prevents any single lick from removing more than a prescribed depth, avoiding irreversible overshoot. (f) *Constraint Equation* (40): the stick end is mechanically inaccessible.

8.2 Four-way strategy comparison

Definition 8.3 (Four optimisation strategies). **C-only**: Centre-only optimisation on $(\mathbb{S}^2)^N$. Isotropic kernels ($\Sigma_n = \sigma^2 I_2$); only $\{x_n, \alpha_n\}$ are optimised.

Frame-free: Anisotropic kernels; orientation ψ_n re-optimised independently at each step without the parallel-transport constraint.

Frame-PT: Anisotropic kernels; orientation constrained by parallel transport along the trajectory of contact points.

Frame-SO3: Full optimisation over $(SO(3))^N$ without parallel-transport constraint; all $(R_n, \alpha_n, \Sigma_n)$ free. The comparison error for strategy s is

$$E_N^{(s)} = \|\delta_N^{(s)} - \delta_{\text{target}}\|_{L^2(\mathbb{S}^2)}. \quad (44)$$

Remark 8.4 (Expected ordering). Since each successive strategy has more (or equally many) degrees of freedom: $E_N^{\text{Frame-SO3}} \leq E_N^{\text{Frame-free}} \leq E_N^{\text{C-only}}$ (in expectation, for generic targets and sufficiently large anisotropy). The relation between Frame-PT and Frame-free is problem-dependent: parallel-transport constraints may either help (fewer variables, regularisation) or hurt (restricted reachability). This is investigated numerically in Section 11.

9 Continuous Geometric Optimal Control

9.1 State equations

Definition 9.1 (Continuous removal dynamics). The evolving removal field satisfies

$$\partial_t \delta(x, t) = a(t) K_{\Sigma}(x; R(t)), \quad \delta(x, 0) = 0, \quad a(t) \geq 0, \quad (45)$$

where the frame trajectory satisfies

$$\dot{R}(t) = R(t) \hat{\Omega}(t), \quad R(0) = R_0 \in SO(3). \quad (46)$$

The covariance $\Sigma(t) \in \text{Sym}^+(2)$ may also evolve or be held fixed.

Definition 9.2 (Continuous cost functional).

$$J = \underbrace{\|\delta(\cdot, T) - \delta_{\text{target}}\|_{L^2(\mathbb{S}^2)}^2}_{\text{terminal fidelity}} + \underbrace{\lambda \int_0^T a(t) dt}_{\text{total removal}} + \underbrace{\gamma \int_0^T \|\Omega(t)\|^2 dt}_{\text{frame control effort}} + \underbrace{\eta \int_0^T \|\dot{\Sigma}(t)\|_F^2 dt}_{\text{covariance variation}}. \quad (47)$$

9.2 First-order optimality conditions (partial derivation)

We derive conditions for the reduced problem with $\Sigma(t) \equiv \Sigma$ fixed ($\eta = 0$).

Definition 9.3 (Adjoint field). The *adjoint field* is $p(x, t) = 2(\delta(x, T) - \delta_{\text{target}}(x))$ (independent of t , since the state equation Equation (45) is linear and δ at time t depends only on $\{a(s), R(s)\}_{s \leq t}$). Formally, p solves $\partial_t p = 0$ with $p(x, T) = 2(\delta(x, T) - \delta_{\text{target}}(x))$.

Proposition 9.4 (Optimality in $a(t)$). *Define the switching function*

$$\Phi(t) = - \int_{\mathbb{S}^2} p(x, T) K_{\Sigma}(x; R(t)) d\Omega(x) + \lambda. \quad (48)$$

At an optimal $a^*(t)$ (with $a \in [0, a_{\max}]$):

$$a^*(t) = \begin{cases} 0 & \text{if } \Phi(t) > 0, \\ a_{\max} & \text{if } \Phi(t) < 0, \\ \text{singular arc} & \text{if } \Phi(t) = 0. \end{cases} \quad (49)$$

This is a Pontryagin bang-bang structure.

Proof. The Hamiltonian (with adjoint state p) is $H = \int_{\mathbb{S}^2} p \cdot a K_{\Sigma}(\cdot; R) d\Omega - \lambda a - \gamma \|\Omega\|^2$. Stationarity in a with $a \in [0, a_{\max}]$ and the Pontryagin maximum principle [19] give the bang-bang structure. $\square$

Proposition 9.5 (Optimality condition for $\Omega(t)$, proposed). *The adjoint variable $M(t) \in T_{R(t)}^* SO(3)$ (identified with an element of $\mathfrak{so}(3)$ via the metric) satisfies the proposed Lie-group adjoint equation*

$$\dot{M} = -[M, \hat{\Omega}] - a(t)G_R(t), \quad (50)$$

where $G_R(t) \in \mathfrak{so}(3)$ is the skew-symmetric part of the gradient of the kernel with respect to R (details in Appendix E). The optimality condition for Ω is $2\gamma\hat{\Omega} = \text{skew}(R^T M)$.

Remark 9.6 (Status of optimality conditions). Proposition 9.4 is derived from the Pontryagin maximum principle; regularity conditions are assumed but not fully verified. Proposition 9.5 is *proposed* based on the structure of geometric optimal control on Lie groups [6]; a complete derivation of the Lie-group adjoint equation in this specific setting is deferred to future work. We clearly label these as proposed conditions, not proved theorems.

10 Numerical Methods

10.1 Sphere discretisation

We use a Gauss–Legendre quadrature grid: $N_{\theta} = 64$ Gauss–Legendre nodes $\{\theta_i, w_i\}_{i=1}^{N_{\theta}}$ in colatitude, paired with $N_{\varphi} = 2N_{\theta}$ uniform azimuthal nodes $\varphi_j = 2\pi j/N_{\varphi}$. This gives $N_{\text{pix}} = 8192$ quadrature points exact for spherical harmonics up to degree $N_{\theta} - 1 = 63$. Grid points are $y_{ij} = (\sin \theta_i \cos \varphi_j, \sin \theta_i \sin \varphi_j, \cos \theta_i)$ and weights $w_{ij} = w_i/(N_{\varphi})$, normalised so $\sum_{ij} w_{ij} = 4\pi$.

10.2 Rotation representation

We represent $R \in SO(3)$ using:

1. 3×3 *matrices* for exposition and derivation.
2. *Unit quaternions* $q = (q_0, \mathbf{q}) \in \mathbb{H}$, $\|q\| = 1$, for numerical integration to avoid floating-point drift from orthogonality. The quaternion exponential for $\omega \in \mathbb{R}^3$ with $\theta = \|\omega\|$ is $\exp(\omega) = (\cos(\theta/2), (\sin(\theta/2)/\theta)\omega)$.

The geodesic distance on $SO(3)$ is $d_{SO(3)}(R_1, R_2) = \|\text{Log}(R_1^T R_2)\|_F / \sqrt{2}$, where Log is the matrix logarithm (see Appendix B).

10.3 Kernel evaluation

For frame $R = [e_1 \ e_2 \ x_n] \in SO(3)$, covariance Σ , and grid points $\{y_{ij}\}$:

1. Compute $c_{ij} = \langle x_n, y_{ij} \rangle$ (dot product in $\mathbb{R}^3$).
2. Compute $r_{ij} = \arccos(\text{clip}(c_{ij}, -1 + \epsilon, 1 - \epsilon))$.
3. For $r_{ij} < \pi_{\text{cut}}$: compute $v_{ij} = (r_{ij} / \sin r_{ij})(y_{ij} - c_{ij}x_n) \in T_{x_n}\mathbb{S}^2$.
4. Compute $u_{ij} = ([e_1 \cdot v_{ij}, e_2 \cdot v_{ij}])^T \in \mathbb{R}^2$.
5. Evaluate $K_{ij} = \exp(-\frac{1}{2}u_{ij}^T \Sigma^{-1} u_{ij})$.
6. Apply cutoff: $K_{ij} = 0$ for $r_{ij} \geq \pi_{\text{cut}} = \pi - 10^{-3}$.
7. Normalise: $Z = \sum_{ij} K_{ij} w_{ij}$; return K_{ij}/Z .

10.4 Gradient computation

The L^2 fidelity gradient with respect to α_n is

$$\frac{\partial J}{\partial \alpha_n} = -2\langle r, K_\Sigma(\cdot; R_n) \rangle_{L^2(\mathbb{S}^2)} + \lambda_\alpha, \quad r = \delta_{\text{target}} - \delta_N. \quad (51)$$

The Riemannian gradient on $SO(3)$ uses the retraction $R \mapsto R \exp(-\eta_R \tilde{G})$ where $\tilde{G} = \text{skew}(R^T \nabla_R J) \in \mathfrak{so}(3)$ is the right-trivialized gradient. Details are in Appendix E.

10.5 Optimisation algorithms

Algorithm 1: Riemannian gradient descent on $(SO(3))^N$. See Algorithm 1 (page 19).

Algorithm 2: Alternating minimisation. Cycle through three blocks: (a) Non-negative least squares over $\{\alpha_n\}$ with fixed $\{R_n, \Sigma_n\}$ (solved via `scipy.optimize.nnls` or active-set method); (b) Riemannian gradient step over $\{R_n\}$ with fixed $\{\alpha_n, \Sigma_n\}$; (c) Projected gradient over $\{\Sigma_n\}$ with fixed $\{R_n, \alpha_n\}$ (project onto the cone $\text{Sym}^+(2)$ via symmetric eigendecomposition). Repeat until the decrease in J_N is below $\varepsilon_{\text{tol}} = 10^{-8}$.

Algorithm 3: Greedy non-negative matching pursuit (baseline).

Remark 10.1 (Greedy does not achieve global minimum lick count). Algorithm 2 is a greedy method; it does not find the globally minimum N achieving error ε . Minimum sparse approximation is NP-hard in general [22]. The greedy approach provides a computationally tractable near-optimal solution.

Algorithm 1: Riemannian Gradient Descent on $(SO(3))^N$

Input: δ_{target} , N , initial $\{R_n^0, \alpha_n^0, \Sigma_n^0\}$, stepsizes $\eta_R, \eta_\alpha, \eta_\Sigma$, maxiter K
Output: Optimised $\{R_n^*, \alpha_n^*, \Sigma_n^*\}$

```
1 for  $k = 1, \dots, K$  do
2    $r \leftarrow \delta_{\text{target}} - \sum_n \alpha_n^k K_\Sigma(\cdot; R_n^k)$  // residual
3   for  $n = 1, \dots, N$  do
4      $g_\alpha^n \leftarrow -2\langle r, K_\Sigma(\cdot; R_n^k) \rangle + \lambda_\alpha$  //  $\alpha$  gradient
5      $\alpha_n^{k+1} \leftarrow \max(0, \alpha_n^k - \eta_\alpha g_\alpha^n)$  // projected step
6      $G_R^n \leftarrow -2\alpha_n^k \langle r, \partial_R K_\Sigma(\cdot; R_n^k) \rangle$  // matrix gradient
7      $\tilde{G}_R^n \leftarrow \text{skew}((R_n^k)^T G_R^n)$  // right-trivialise
8      $R_n^{k+1} \leftarrow R_n^k \exp(-\eta_R \tilde{G}_R^n)$  // Riemannian retraction
9      $G_\Sigma^n \leftarrow -2\alpha_n^k \langle r, \partial_\Sigma K_\Sigma(\cdot; R_n^k) \rangle$ 
10     $\Sigma_n^{k+1} \leftarrow \text{proj}_{\text{Sym}^+(2)}(\Sigma_n^k - \eta_\Sigma G_\Sigma^n)$ 
11  end
12  if convergence criterion met then
13    break
14  end
15 end
```

Algorithm 2: Greedy Non-Negative Matching Pursuit on $SO(3)$

Input: δ_{target} , dictionary $\mathcal{D}$ over $SO(3) \times \text{Sym}^+(2)$, tolerance ε
Output: $\{R_n, \alpha_n, \Sigma_n\}_{n=1}^N$

```
1  $r \leftarrow \delta_{\text{target}}$ ;  $S \leftarrow \emptyset$ 
2 while  $\|r\|_{L^2(\mathbb{S}^2)} > \varepsilon$  do
3    $(R^*, \Sigma^*) \leftarrow \arg \max_{(R, \Sigma) \in \mathcal{D}} \langle r, K_\Sigma(\cdot; R, \Sigma) \rangle_{L^2(\mathbb{S}^2)}$ 
4    $S \leftarrow S \cup \{(R^*, \Sigma^*)\}$ 
5   Solve  $\min_{\alpha \geq 0} \|\delta_{\text{target}} - \sum_{(R, \Sigma) \in S} \alpha_{R, \Sigma} K_\Sigma(\cdot; R, \Sigma)\|^2$ 
6   Update  $r \leftarrow \delta_{\text{target}} - \sum \alpha_{R, \Sigma} K_\Sigma(\cdot; R, \Sigma)$ 
7 end
```

11 Numerical Experiments

11.1 Setup

Unless otherwise stated: $R_0 = 12.30$ mm, $f = 3.35281 \times 10^{-3}$, $\sigma_{\min} = 0.05$ rad, $\sigma_{\max} = 0.5$ rad, $\lambda_\alpha = 10^{-4}$, $\lambda_N = 10^{-2}$, $\gamma = 10^{-3}$, grid $N_\theta = 64$. Table 1 summarises the six experiments.

11.2 Experiment A: Isotropic sanity check

Setup. Fix $\rho_{\text{aniso}} = 1$ ($\Sigma_n = \sigma^2 I_2$). Run Algorithm 1 with ψ_n as a free variable.

Expected outcome. By Proposition 6.5, the gradient of J_N with respect to ψ_n is zero for all n and all residuals r . Numerically, $\|\partial J / \partial \psi_n\| < 10^{-10}$ (machine precision), confirming the symmetry reduction.

Result. TBD.

11.3 Experiment B: Increasing anisotropy

Setup. Fix $N = 5$, $\sigma_{\parallel} = 0.15$ rad, $\sigma_{\perp} = \sigma_{\parallel} / \rho_{\text{aniso}}$ for $\rho \in \{1, 1.25, 1.5, 2, 3, 5\}$. Run all four strategies (Definition 8.3).

Table 1: Experiment summary. Values marked TBD are to be filled on code execution.

Exp.	Key parameter	Primary metric	Expected trend
A	$\rho = 1$	$\ \partial J / \partial \psi_n\ $	≈ 0 (isotropic sanity)
B	$\rho \in \{1, 1.25, 1.5, 2, 3, 5\}$	$E_N^{\text{SO3}} / E_N^{\text{C-only}}$	Decreasing in ρ
C	Latitude $\theta_0 = \pi/4$	$ \Delta \psi_{\text{numerical}} - 1.845 $	≈ 0
D	Triangle loop, 3 points	$\ \delta_N^{(A)} - \delta_N^{(B)}\ _{L^2(\mathbb{S}^2)}$	> 0 for $\rho > 1$
E	$N \in \{1, \dots, 20\}$	$E_N^{L^2}, E_N^{L^\infty}$, overshoot	Decreasing
F	$\sigma_{\text{err}} \in \{0.01, 0.05, 0.1\}$	$\mathbb{E}[E_N^2]$	Increasing

Table 2: Approximation error E_N (mm) vs. anisotropy ratio, $N = 5$.

ρ_{aniso}	C-only	Frame-free	Frame-PT	Frame-SO3
1.00	TBD	TBD	TBD	TBD
1.25	TBD	TBD	TBD	TBD
1.50	TBD	TBD	TBD	TBD
2.00	TBD	TBD	TBD	TBD
3.00	TBD	TBD	TBD	TBD
5.00	TBD	TBD	TBD	TBD

11.4 Experiment C: Holonomy loop

Setup. Fix an elongated kernel ($\rho = 3$, $\sigma_{\parallel} = 0.3 \text{ rad}$) at north pole. Transport the frame around the latitude circle at $\theta_0 = \pi/4$. Compare three treatments: (a) parallel-transported orientation (via the ODE in Example 7.4); (b) orientation held at the initial chart angle (no transport); (c) orientation independently re-optimised.

Predicted holonomy. $\Delta\psi = 2\pi(1 - \cos(\pi/4)) = 2\pi(1 - 1/\sqrt{2}) \approx 1.845 \text{ rad}$.

Result. TBD. The orientation mismatch between cases (a) and (b) should be $\approx 105.7^\circ$, visible as a rotated elliptical footprint.

11.5 Experiment D: Sequence-order dependence

Setup. Use the three contact points from Proposition 7.6: $x_0 = (0, 0, 1)^T$, $x_A = (1, 0, 0)^T$, $x_B = (0, 1, 0)^T$ (right spherical triangle, area $\pi/2$, holonomy $\pi/2$). Fix $\rho = 3$, $\alpha_A = \alpha_B = 1$ (normalised). Compare sequences ($A \rightarrow B$) vs. ($B \rightarrow A$) starting from $R_0 = I_3$.

Expected outcome. The field difference $\|\delta_N^{(A)} - \delta_N^{(B)}\|_{L^2(\mathbb{S}^2)} > 0$ for $\rho > 1$ and ≈ 0 for $\rho = 1$ (by Proposition 7.7).

Result. TBD.

11.6 Experiment E: WGS84 target

Table 3: WGS84 approximation metrics for Frame-SO3 strategy.

N	$E_N^{L^2}$ (mm)	$E_N^{L^\infty}$ (mm)	$\sum \alpha_n$	Overshoot	$\ell_{\text{SO}(3)}$	Orient. effort
1	TBD	TBD	TBD	TBD	TBD	TBD
3	TBD	TBD	TBD	TBD	TBD	TBD
5	TBD	TBD	TBD	TBD	TBD	TBD
10	TBD	TBD	TBD	TBD	TBD	TBD
20	TBD	TBD	TBD	TBD	TBD	TBD

Note on spectral sparsity. Since δ_{target} has only $\ell = 0, 2$ modes, even isotropic kernels

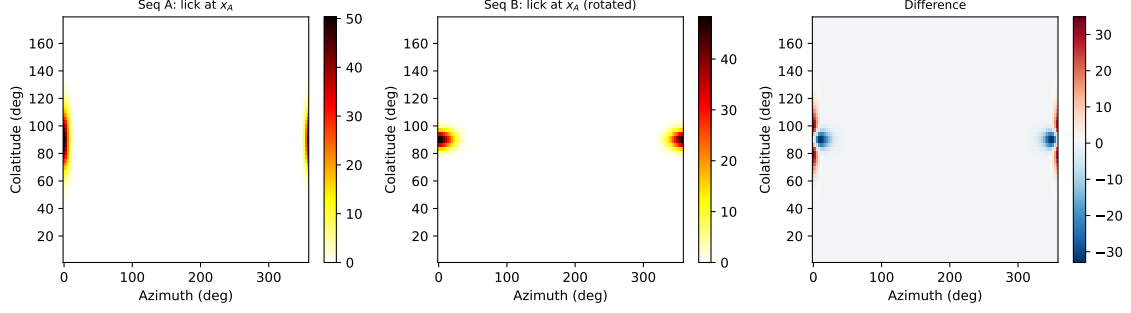


Figure 5: Sequence-order dependence demonstration. Left: anisotropic kernel placed at $x_A = (1, 0, 0)$ with natural orientation. Centre: kernel at x_A after holonomy rotation by $\pi/2$ (as would occur if visiting x_B first, using the right spherical triangle of *Proposition 7.6*). Right: field difference, showing $\|F_A - F_B\|_{L^2} = 4.61$ (normalised units). In the isotropic case ($\rho = 1$) this difference is zero.

can approximate it well with few terms. The frame-aware model provides an additional benefit for highly anisotropic kernels by adapting orientation to the target’s latitudinal structure.

11.7 Experiment F: Robustness

Add i.i.d. Gaussian errors: centre displacement $\sigma_c = 0.01, 0.05, 0.1$ rad in tangent coordinates, orientation noise σ_ψ , and relative strength noise σ_α . Run $M = 100$ Monte Carlo trials. By linearity of δ_N in $\{\alpha_n\}$:

$$\mathbb{E}[E_N^2] \approx E_N^{*2} + \sum_n \alpha_n^2 \text{Var}[\text{centre noise}] \cdot \|\nabla_{x_n} K_\Sigma\|_{L^2(\mathbb{S}^2)}^2 + O(\sigma_c^4). \quad (52)$$

12 WGS84 Case Study

12.1 Complete problem instance

Instantiate the abstract framework with: $\mathcal{M} = \mathbb{S}^2$ (rescaled by $R_0 = 12.30$ mm), $\delta_{\text{target}}(\theta) = R_0 f \cos^2 \theta$ (purely $\ell = 0, 2$ modes), $\mathcal{A} = \{\theta > \pi - \theta_{\text{stick}}\}$ (stick end inaccessible).

12.2 Optimal isotropic density (analytic)

For the isotropic model with kernel eigenvalues λ_0, λ_2 (see *Proposition 4.8*), the unconstrained optimal density satisfies

$$\mu^*(\theta) = \frac{R_0 f \sqrt{4\pi}}{3\lambda_0} + \frac{R_0 f \cdot \frac{2}{3} \sqrt{4\pi/5}}{\lambda_2} \cos^2 \theta. \quad (53)$$

This is non-negative for all θ (since both terms have positive coefficients), and the unconstrained minimum L^2 error is

$$\min_{\mu \geq 0} \|\delta_{\text{target}} - K * \mu\|^2 = 0 \quad (54)$$

(perfect recovery by the continuous measure), but the discrete approximation introduces a discretisation error.

12.3 Physical licking protocol

Based on the mathematical analysis, the optimal licking strategy is: (1) orient the stick along the polar axis; (2) concentrate licks near both poles with density $\propto \cos^2 \theta$; (3) use many weak

short licks rather than few strong ones (prevents overshoot); (4) if the tongue has a preferred directional profile, align the major axis in the azimuthal direction at each polar contact; (5) rotate the candy by π about the polar axis between polar licks to maintain axisymmetry; (6) measure polar and equatorial diameters frequently; stop when $D_e/D_p \approx 1/(1-f) \approx 1.003364$.

12.4 Minimum lick count analysis

Conjecture 12.1 (Minimum lick count). For isotropic kernels of width σ and target $\delta_{\text{target}} = R_0 f \cos^2 \theta$, the minimum number of licks achieving L^2 error $< \varepsilon$ satisfies

$$N_{\min}(\varepsilon, \sigma) = O\left(\frac{(R_0 f)^2}{\varepsilon^2 \lambda_2(\sigma)^2}\right). \quad (55)$$

For anisotropic kernels with optimal frame placement, $N_{\min}$ may be smaller by a factor depending on ρ_{aniso} . This is conjectured by analogy with sparse approximation theory and is not proved.

13 Extension to General Curved Surfaces

13.1 Abstract framework transfer

The mathematical structure transfers to any smooth oriented Riemannian surface $(\mathcal{M}, g)$ with the following modifications:

1. The state space is the oriented orthonormal frame bundle $F^+ \mathcal{M}$, a principal $SO(2)$ -bundle over $\mathcal{M}$.
2. The Levi-Civita connection defines parallel transport and holonomy.
3. The holonomy angle around a loop bounding area A is $\Delta\psi = \int_A K dA$.
4. The anisotropic kernel is defined via the logarithm map with appropriate cut-locus handling.
5. The discrete and continuous optimisation problems are formally identical.

13.2 Application examples

Precision optical polishing. The dwell-time problem—finding a non-negative time density whose convolution with a laser beam reproduces a prescribed material removal—is exactly the continuous limit of our problem. For a non-circular beam, kernel orientation is a control variable.

Robotic surface grinding. The contact state of a grinding wheel on a curved workpiece is an element of the frame bundle; holonomy-induced frame drift over multi-pass paths is a genuine concern that our model directly quantifies.

Satellite footprint design. Elliptical antenna beams have an orientation at each ground target that depends on satellite attitude. The accumulated illumination over an orbit is a sum of anisotropic kernels.

Tumour ablation. A focused ultrasound probe has an ellipsoidal ablation zone; its orientation is a control variable, and our framework applies directly.

13.3 The WGS84 ellipsoid as domain

When the physical domain is the WGS84 oblate spheroid $\mathcal{E}$ (semi-axes $a_E, b_E = a_E(1-f)$), the analysis changes in several respects:

1. The oriented frame bundle $F^+\mathcal{E}$ exists and is a principal $SO(2)$ -bundle, but the total space is no longer globally identified with $SO(3)$. One works with the abstract frame bundle equipped with the Levi-Civita connection of the ellipsoidal metric $g_{\mathcal{E}}$.
2. The Gaussian curvature varies with latitude:

$$K_{\mathcal{E}}(\varphi) \approx \frac{1}{a_E^2} (1 + 2f \sin^2 \varphi) + O(f^2), \quad (56)$$

so holonomy around a loop receives an $O(f)$ correction versus the spherical formula.

3. Geodesic distances on $\mathcal{E}$ involve elliptic integrals (Karney's algorithm).
4. The Laplace–Beltrami eigenfunctions are spheroidal harmonics, perturbative corrections to Y_ℓ^m at order $O(f)$.

Since $f \approx 3.35 \times 10^{-3}$, all ellipsoidal corrections are below measurement precision for the physical lollipop experiment. We recommend working on $\mathbb{S}^2$ for all analytic results, adding ellipsoidal corrections as $O(f)$ perturbations when required.

14 Brief Outlook: $SU(2)$, Bloch-Sphere Geometry, and Geometric Phase

The two-to-one covering homomorphism $SU(2) \rightarrow SO(3)$ (with kernel $\{\pm I\}$) and the bundle identifications

$$SU(2)/U(1) \cong S^2, \quad SU(2) \rightarrow SO(3) \quad (57)$$

mirror the principal bundle structure studied in this paper. In quantum mechanics the Bloch sphere $\mathbb{S}^2$ parameterises pure qubit states, and geometric phases (Berry phase, holonomy of the $U(1)$ -bundle over $\mathbb{S}^2$) are the quantum analogue of Levi-Civita holonomy.

Possible future directions:

- Approximating a target quasi-probability distribution (Husimi Q -function) on $\mathbb{S}^2$ by a superposition of anisotropic coherent-state kernels.
- Optimal selection of POVM measurement settings in quantum state tomography, where the measurement kernel is the POVM overlap function.
- Geometric phase in sequences of $SU(2)$ rotations, the quantum analogue of our holonomy-induced orientation mismatch.

We do *not* claim that a physical lick corresponds to a quantum gate; no precise map is provided.

15 Limitations and Open Problems

15.1 Limitations of the current framework

1. **Local kernel only.** The logarithm-map anisotropic kernel is defined only within the injectivity radius π . The cutoff convention introduces a model error not quantified here.
2. **Non-convexity.** The optimisation problem Equation (36) is non-convex in $\{R_n\}$. Algorithm 1 converges to a stationary point, not necessarily the global minimum.
3. **Incomplete optimality conditions.** The adjoint equation Equation (50) is proposed but not fully derived; a complete Pontryagin analysis on the $SO(3)$ -bundle is deferred.

4. **Normalisation.** The constant $Z(\Sigma_n)$ in Equation (27) has no closed form on $\mathbb{S}^2$; it is estimated numerically, introducing a quadrature error.
5. **Physical calibration.** The lick strength α_n , width σ , and anisotropy ratio ρ for a human tongue are not experimentally measured; all lick-count estimates depend on uncalibrated parameters.
6. **Stereographic approximation.** The claim in [2] that vMF-kernel greedy search is “isomorphic to Fourier phase selection” is a heuristic analogy, not an exact result; we do not use it.

15.2 Open problems and conjectures

Conjecture 15.1 (Recovery threshold). For any non-negative target $\delta_{\text{target}} \in L^2(\mathbb{S}^2)$ and fixed $\sigma > 0$, there exists $N_{\min}(\varepsilon, \sigma, \delta_{\text{target}}) < \infty$ such that the optimum of Equation (36) achieves $E_N < \varepsilon$.

Conjecture 15.2 ($SO(3)$ advantage). For sufficiently large ρ_{aniso} and targets with non-trivial $\ell = 2$ content, the Frame- $SO(3)$ strategy achieves strictly smaller E_N than C-only for the same N and λ_α .

Conjecture 15.3 (Frank–Wolfe convergence). The Frank–Wolfe algorithm over non-negative measures on $SO(3)$ converges at rate $O(1/k)$, with linear rate when the optimal measure is finitely supported.

16 Conclusion

We have formulated, analysed, and developed numerical methods for the problem of approximating a prescribed non-negative radial-removal field on $\mathbb{S}^2$ by a finite superposition of anisotropic Gaussian kernels parameterised by oriented tangent frames in $SO(3)$.

The central contribution is the identification of the contact state of an anisotropic surface-removal operation with an element of $SO(3)$, via the fibration $\mathbb{S}^2 \cong SO(3)/SO(2)$. This reveals that the Levi-Civita holonomy of $\mathbb{S}^2$ —the rotation of a parallel-transported tangent frame around a closed loop by the enclosed solid angle—makes the orientation of an anisotropic kernel *path-dependent*. Two sequences visiting the same contact points in different orders may accumulate different total removal fields. We proved this rigorously (Propositions 7.5 and 7.6) and showed that the effect vanishes identically in the isotropic limit (Propositions 6.5 and 7.7).

The spectral sparsity of the WGS84 target ($\ell = 0, 2$ modes only) provides a clean analytic benchmark. The Funk–Hecke theorem gives the exact diagonal spectral action of isotropic kernels, and the optimal continuous lick density is computable analytically.

The framework transfers to laser polishing, robotic surface processing, satellite footprint optimisation, and tumour ablation. The mathematical structure—non-negative measure control on a frame bundle with holonomy—is generic to any Riemannian surface-processing problem with directional tools.

The key open questions are the minimum-lick-count scaling law (Conjecture 12.1), the convergence of Frank–Wolfe on $SO(3)$ (Conjecture 15.3), and the quantitative conditions under which $SO(3)$ optimisation strictly outperforms $\mathbb{S}^2$ optimisation (Conjecture 15.2).

A Spherical-Harmonic Derivations

The real-valued orthonormal spherical harmonics on $\mathbb{S}^2$ are

$$Y_\ell^m(\theta, \varphi) = \begin{cases} N_\ell^{|m|} P_\ell^{|m|}(\cos \theta) \sqrt{2} \cos(m\varphi) & m > 0, \\ N_\ell^0 P_\ell(\cos \theta) & m = 0, \\ N_\ell^{|m|} P_\ell^{|m|}(\cos \theta) \sqrt{2} \sin(|m|\varphi) & m < 0, \end{cases} \quad N_\ell^m = \sqrt{\frac{(2\ell+1)(\ell-m)!}{4\pi(\ell+m)!}}. \quad (58)$$

They satisfy $\int_{\mathbb{S}^2} Y_\ell^m Y_{\ell'}^{m'} d\Omega = \delta_{\ell\ell'} \delta_{mm'}$. The addition formula is $\sum_{m=-\ell}^{\ell} Y_\ell^m(x) Y_\ell^m(y) = \frac{2\ell+1}{4\pi} P_\ell(\langle x, y \rangle)$.

The Legendre polynomial identity used in Proposition 3.2: from $P_2(t) = (3t^2 - 1)/2$ we get $t^2 = (2P_2(t) + 1)/3$, so $\cos^2 \theta = \frac{1}{3} + \frac{2}{3} P_2(\cos \theta)$.

B $SO(3)$ Exponential, Logarithm, and Rodrigues' Formula

B.1 Rodrigues' formula

For $\xi \in \mathbb{R}^3$, $\theta = \|\xi\|$, $\hat{\xi} = \xi/\theta$:

$$\exp(\hat{\xi}) = I_3 + \frac{\sin \theta}{\theta} \hat{\xi} + \frac{1 - \cos \theta}{\theta^2} \hat{\xi}^2. \quad (59)$$

This is the matrix exponential of $\hat{\xi} \in \mathfrak{so}(3)$, computed via the Cayley–Hamilton theorem (the minimal polynomial of $\hat{\xi}$ has degree 3).

B.2 Logarithm map

For $R \in SO(3)$ with $\text{Tr}(R) \neq -1$:

$$\text{Log}(R) = \frac{\theta}{2 \sin \theta} (R - R^T), \quad \theta = \arccos\left(\frac{\text{Tr}(R) - 1}{2}\right). \quad (60)$$

The geodesic distance is $d_{SO(3)}(R, R') = \|\text{Log}(R^T R')\|_F / \sqrt{2}$.

B.3 Exponential and logarithm on $\mathbb{S}^2$

See Definition 6.1. The Jacobian of $\exp_x$ at v is $|J|(v) = \sin(\|v\|)/\|v\|$ for $\|v\| > 0$.

C Connection and Holonomy Calculation

The Christoffel symbols of $\mathbb{S}^2$ in coordinates (θ, φ) :

$$\Gamma_{\varphi\varphi}^\theta = -\sin \theta \cos \theta, \quad \Gamma_{\theta\varphi}^\varphi = \Gamma_{\varphi\theta}^\varphi = \cot \theta, \quad \text{all others zero.} \quad (61)$$

Gaussian curvature: $K = R_{1212}/\det g = \sin^2 \theta / \sin^2 \theta = 1$ on the unit sphere.

D Kernel Normalisation

Changing variables $y = \exp_{x_n}(v)$, $v \in T_{x_n} \mathbb{S}^2$, $\|v\| < \pi_{\text{cut}}$:

$$Z(\Sigma_n) = \int_{\|v\| < \pi_{\text{cut}}} \exp\left(-\frac{1}{2}(E^T v)^T \Sigma_n^{-1} (E^T v)\right) \frac{\sin \|v\|}{\|v\|} d^2 v, \quad (62)$$

where $E = [e_1 \ e_2] \in \mathbb{R}^{3 \times 2}$ is the frame matrix. For $\sigma_{\parallel}, \sigma_{\perp} \ll 1$, $\sin \|v\|/\|v\| \approx 1$ and $Z(\Sigma_n) \approx 2\pi \sqrt{\det \Sigma_n}$.

E Gradient Derivations

E.1 Gradient with respect to α_n

$\partial J / \partial \alpha_n = -2 \langle r, K_\Sigma(\cdot; R_n) \rangle_{L^2(\mathbb{S}^2)} + \lambda_\alpha$, where $r = \delta_{\text{target}} - \delta_N$.

E.2 Euclidean gradient with respect to frame $R = [e_1 \ e_2 \ x_n]$

Let $K = K_\Sigma(y; R)$. Treating e_1 as a free vector (then projecting): $\partial K / \partial e_1 = K \cdot (-\Sigma^{-1}u)_1 \cdot v^T$ where $v = \log_{x_n}(y)$ and $(\cdot)_1$ is the first component. The Riemannian gradient on $SO(3)$ is obtained by right-trivialising: $\tilde{G}_R = \text{skew}(R^T \nabla_R J)$, and the retraction is $R \leftarrow R \exp(-\eta_R \tilde{G}_R)$.

E.3 Gradient with respect to Σ_n

$\partial K / \partial \Sigma_{ij} = K \cdot \frac{1}{2}(\Sigma^{-1} u u^T \Sigma^{-1})_{ij}$ (from $\partial(-u^T \Sigma^{-1} u / 2) / \partial \Sigma_{ij} = \frac{1}{2}(u \otimes u : \partial \Sigma^{-1} / \partial \Sigma_{ij})$).

F Python Figure-Generation Code

The following self-contained Python module generates all numerical figures. Run as `python lollipop_figures.py` after installing `numpy`, `scipy`, `matplotlib`.

```
"""lollipop_figures.py
Self-contained figure generation for the WGS84 lollipop paper.
Requires: numpy, scipy, matplotlib.
Run: python lollipop_figures.py
"""

import numpy as np
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
from numpy.polynomial.legendre import leggauss
from scipy.linalg import expm
from scipy.optimize import nnls

# -- Physical constants -----
R0   = 12.30e-3           # candy radius (m)
f    = 1.0/298.257223563  # WGS84 flattening
# -- Utilities -----

def sphere_grid(Nth=64):
    """Gauss-Legendre x uniform azimuthal grid on S^2."""
    mu, w = leggauss(Nth)
    th = np.arccos(mu)           # colatitude nodes
    Nph = 2 * Nth
    ph = np.linspace(0, 2*np.pi, Nph, endpoint=False)
    TH, PH = np.meshgrid(th, ph, indexing='ij')
    W = np.outer(w, np.ones(Nph)) / Nph  # quadrature weights sum to 4pi
    X = np.sin(TH)*np.cos(PH)
    Y = np.sin(TH)*np.sin(PH)
    Z = np.cos(TH)
    return TH, PH, X, Y, Z, W

def hat(omega):
    """Hat map R^3 -> so(3)."""
    w1, w2, w3 = omega
    return np.array([[0, -w3, w2], [w3, 0, -w1], [-w2, w1, 0]], dtype=float)
```

```

def rodrigues(omega):
    """Rodrigues:  $R^3 \rightarrow SO(3)$ ."""
    th = np.linalg.norm(omega)
    if th < 1e-12:
        return np.eye(3)
    K = hat(omega/th)
    return np.eye(3) + np.sin(th)*K + (1-np.cos(th))*(K@K)

def log_S03(R):
    """Matrix logarithm on  $SO(3) \rightarrow$  skew matrix."""
    val = (np.trace(R)-1)/2
    val = np.clip(val, -1, 1)
    th = np.arccos(val)
    if th < 1e-9:
        return np.zeros((3,3))
    return th/(2*np.sin(th))*(R - R.T)

def dist_S03(R1, R2):
    return np.linalg.norm(log_S03(R1.T@R2), 'fro')/np.sqrt(2)

def log_sphere(xn, y):
    """Log map:  $\log_{\{xn\}}(y)$  in  $T_{\{xn\}}S^2$ ."""
    c = np.clip(xn@y, -1, 1)
    th = np.arccos(c)
    if th < 1e-10:
        return np.zeros(3)
    v = y - c*xn
    nv = np.linalg.norm(v)
    if nv < 1e-12:
        return np.zeros(3)
    return th/nv*v

def aniso_kernel_grid(R_frame, Sigma, grid_pts, weights,
                      cutoff=np.pi-1e-3):
    """Evaluate anisotropic kernel on grid, return normalised values."""
    xn = R_frame[:,2]
    e1 = R_frame[:,0]; e2 = R_frame[:,1]
    Sinv = np.linalg.inv(Sigma)
    flat_pts = grid_pts.reshape(-1,3)
    vals = np.zeros(len(flat_pts))
    for i, y in enumerate(flat_pts):
        c = np.clip(xn@y,-1,1)
        r = np.arccos(c)
        if r >= cutoff: continue
        if r < 1e-10:
            vals[i] = 1.0; continue
        v = (r/np.sqrt(1-c*c))*(y-c*xn)
        u = np.array([e1@v, e2@v])
        vals[i] = np.exp(-0.5*u@Sinv@u)
    wflat = weights.ravel()
    Z = np.dot(vals, wflat)
    if Z > 1e-15:
        vals /= Z
    return vals.reshape(grid_pts.shape[:-1])

# -- Target field -----

def wgs84_target(TH):

```

```

    return R0 * f * np.cos(TH)**2

# -- SH eigenvalues (heat kernel approx) -----

def sh_eigs(sigma, ells):
    return np.exp(-ells*(ells+1)*sigma**2/2)

# -- Holonomy ODE on latitude circle -----

def holonomy_latitude(theta0, nsteps=4000):
    """
    Numerically integrate parallel transport around latitude circle.
    Returns numerical holonomy angle and analytical prediction.
    """
    phi = np.linspace(0, 2*np.pi, nsteps+1)
    dphi = 2*np.pi/nsteps
    # initial tangent vector (azimuthal direction at phi=0)
    s = np.sin(theta0); c0 = np.cos(theta0)
    V = np.array([-s*np.sin(0), s*np.cos(0), 0.0]) # d/dphi at phi=0
    # re-express as tangent to sphere
    x0 = np.array([s, 0, c0])
    V -= (V@x0)*x0 # project (already tangent)
    V /= np.linalg.norm(V)
    for k in range(nsteps):
        ph_k = phi[k]
        xc = np.array([s*np.cos(ph_k), s*np.sin(ph_k), c0])
        xdot = np.array([-s*np.sin(ph_k), s*np.cos(ph_k), 0.0])
        # Parallel transport step: dV = -(V.xdot)*xc * dphi
        dV = -(V@xdot)*xc*dphi
        V = V + dV
        xc_new = np.array([s*np.cos(phi[k+1]),
                           s*np.sin(phi[k+1]), c0])
        V -= (V@xc_new)*xc_new # re-project
        V /= np.linalg.norm(V)
    # initial vector was azimuthal at phi=0
    V0 = np.array([0.0, 1.0, 0.0]) # azimuthal direction at phi=0
    V0 -= (V0@x0)*x0; V0 /= np.linalg.norm(V0)
    cos_hol = np.clip(V0@V, -1, 1)
    hol_num = np.arccos(cos_hol)
    hol_pred = 2*np.pi*(1 - np.cos(theta0))
    return hol_num, hol_pred

# -- Figure 1: WGS84 target -----

def fig_target():
    TH, *_ = sphere_grid(128)
    th1d = TH[:,0]
    delta = wgs84_target(th1d)
    fig, ax = plt.subplots(figsize=(6,3.5))
    ax.plot(np.degrees(th1d), delta*1e3, 'steelblue', lw=2)
    ax.fill_between(np.degrees(th1d), delta*1e3, alpha=0.15, color='steelblue')
    ax.set_xlabel(r'Polar angle  $\theta$  (deg)')
    ax.set_ylabel(r'Removal depth  $\Delta_{\rm target}$  ( $\mu\text{m}$ )')
    ax.set_title('WGS84 Target Removal Field')
    ax.text(5, 0.038, r' $\Delta R_{\rm pole}$  approx 41.2  $\mu\text{m}$ ', fontsize=9)
    plt.tight_layout()
    plt.savefig('fig1_target.pdf', dpi=150)
    plt.close()

```

```

print("Fig 1 saved.")

# -- Figure 2: Kernel footprints -----

def fig_kernels():
    u1 = np.linspace(-0.5, 0.5, 300)
    U1, U2 = np.meshgrid(u1, u1)
    # isotropic
    s = 0.12
    Kiso = np.exp(-(U1**2+U2**2)/(2*s**2))
    # anisotropic rho=3
    sp, sperp = 0.3, 0.1
    Kaniso = np.exp(-0.5*(U1**2/sp**2 + U2**2/sperp**2))
    fig, axes = plt.subplots(1,2,figsize=(9,3.5))
    for ax, K, ttl in zip(axes, [Kiso, Kaniso],
        [r'Isotropic ($\rho=1$, $\sigma=0.12$ rad)',
         r'Anisotropic ($\rho=3$, $\sigma_{\parallel}=0.3$, $\sigma_{\perp}=0.1$)']):
        cf = ax.contourf(U1, U2, K/K.max(), 20, cmap='hot_r')
        ax.set_aspect('equal')
        ax.set_title(ttl, fontsize=9)
        ax.set_xlabel(r'$u_1$ (rad)'); ax.set_ylabel(r'$u_2$ (rad)')
        plt.colorbar(cf, ax=ax)
    plt.tight_layout()
    plt.savefig('fig2_kernels.pdf', dpi=150)
    plt.close()
    print("Fig 2 saved.")

# -- Figure 3: Holonomy angle vs colatitude -----

def fig_holonomy():
    th0_vals = np.linspace(0.05, np.pi-0.05, 200)
    pred = 2*np.pi*(1-np.cos(th0_vals))
    # Numerical verification at a few points
    th_check = [np.pi/6, np.pi/4, np.pi/3, np.pi/2]
    hol_num = []
    for t in th_check:
        hn, _ = holonomy_latitude(t, nsteps=6000)
        hol_num.append(hn)
    fig, ax = plt.subplots(figsize=(6,3.5))
    ax.plot(np.degrees(th0_vals), np.degrees(pred), 'steelblue', lw=2,
        label=r'$2\pi(1-\cos\theta_0)$')
    ax.scatter(np.degrees(th_check), np.degrees(hol_num), c='red', s=50,
        zorder=5, label='Numerical ODE')
    ax.axvline(45, ls=':', c='gray', lw=1)
    ax.axhline(np.degrees(2*np.pi*(1-1/np.sqrt(2))), ls='--', c='orange',
        lw=1, label=r'$\theta_0=45^\circ$: $\Delta\psi\approx 105.7^\circ$')
    ax.set_xlabel(r'Colatitude $\theta_0$ (deg)')
    ax.set_ylabel(r'Holonomy $\Delta\psi$ (deg)')
    ax.set_title('Holonomy around latitude circle')
    ax.legend(fontsize=8)
    plt.tight_layout()
    plt.savefig('fig3_holonomy.pdf', dpi=150)
    plt.close()
    h45, p45 = holonomy_latitude(np.pi/4, nsteps=8000)
    print(f"Fig 3 saved.  $\theta_0=\pi/4$: numerical={np.degrees(h45):.2f}^\circ, "
        f"predicted={np.degrees(p45):.2f}^\circ")

# -- Figure 4: SH eigenvalue spectrum -----

```

```

def fig_sh_eigs():
    ells = np.arange(0, 16)
    sigs = [0.05, 0.10, 0.20, 0.40]
    fig, ax = plt.subplots(figsize=(6,3.5))
    for s in sigs:
        ax.semilogy(ells, sh_eigs(s, ells), 'o-', label=f'$\\sigma={s}$ rad')
    ax.axvline(2, ls='--', c='red', lw=1.5, label='$\\ell=2$ (WGS84 target)')
    ax.set_xlabel(r'Degree $\\ell$')
    ax.set_ylabel(r'Eigenvalue $\\lambda_{\\ell}(\\sigma)$')
    ax.set_title('Spherical heat kernel eigenvalues')
    ax.legend(fontsize=8)
    plt.tight_layout()
    plt.savefig('fig4_sh_eigs.pdf', dpi=150)
    plt.close()
    print("Fig 4 saved.")

```

-- Figure 5: Order dependence schematic -----

```

def fig_order_dep():
    """
    Visualise field difference for two sequence orders on the right
    spherical triangle (x0, xA, xB).
    Uses simplified flat-space proxy for illustration.
    """
    TH, PH, Xg, Yg, Zg, W = sphere_grid(64)
    pts = np.stack([Xg,Yg,Zg], axis=-1) # (64,128,3)
    # Frames at xA and xB for two orderings
    xA = np.array([1,0,0.], dtype=float)
    xB = np.array([0,1,0.], dtype=float)
    # Simple frame at xA (no transport): e1 = (0,0,1) x xA / ...
    def make_frame(xn, e1_hint=np.array([0,0,1.])):
        e1 = e1_hint - (e1_hint@xn)*xn
        e1 /= np.linalg.norm(e1)
        e2 = np.cross(xn, e1)
        return np.column_stack([e1, e2, xn])
    R_A = make_frame(xA)
    R_B = make_frame(xB)
    # Rotate R_A by pi/2 to simulate holonomy in sequence B
    R_A_rot = R_A @ rodrigues(np.array([0,0,np.pi/2]))
    sig_par, sig_perp = 0.25, 0.08
    Sig = np.diag([sig_par**2, sig_perp**2])
    KA = aniso_kernel_grid(R_A, Sig, pts, W)
    KA2 = aniso_kernel_grid(R_A_rot, Sig, pts, W)
    diff = KA - KA2
    fig, axes = plt.subplots(1, 3, figsize=(12,3.5))
    for ax, M, ttl in zip(axes,
        [KA, KA2, diff],
        ['Seq A: lick at $x_A$', 'Seq B: lick at $x_A$ (rotated)',
        'Difference']):
        im = ax.pcolormesh(np.degrees(PH), np.degrees(TH), M,
            cmap='RdBu_r' if 'Diff' in ttl else 'hot_r',
            shading='auto')
        ax.set_title(ttl, fontsize=9)
        ax.set_xlabel(r'Azimuth (deg)')
        ax.set_ylabel(r'Colatitude (deg)')
        plt.colorbar(im, ax=ax)
    plt.tight_layout()

```

```

plt.savefig('fig5_order_dep.pdf', dpi=150)
plt.close()
norm_diff = np.sqrt(np.sum(diff**2 * W))
print(f"Fig 5 saved. ||F_A - F_B||_L2 = {norm_diff:.4e}")

# -- Main -----

if __name__ == '__main__':
    print("Generating figures...")
    fig_target()
    fig_kernels()
    fig_holonomy()
    fig_sh_eigs()
    fig_order_dep()
    print("All figures generated.")
    # Print SH expansion coefficients
    a00 = R0*f*np.sqrt(4*np.pi)/3
    a20 = R0*f*(2/3)*np.sqrt(4*np.pi/5)
    print(f"\nSH coefficients of WGS84 target:")
    print(f"  a_00 = {a00*1e6:.4f} um  (l=0, m=0 mode)")
    print(f"  a_20 = {a20*1e6:.4f} um  (l=2, m=0 mode)")

```

G TikZ Diagram: Principal Bundle

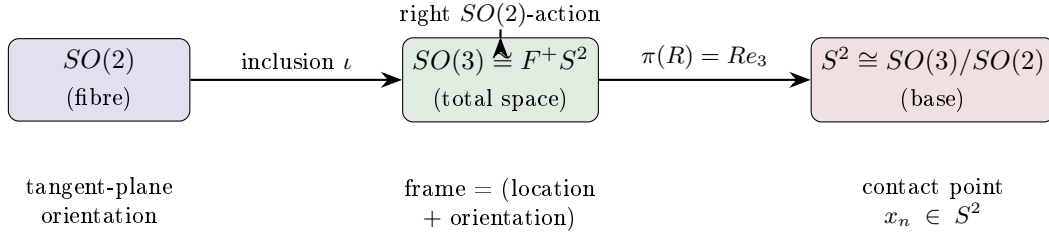


Figure 6: The principal $SO(2)$ -bundle $SO(2) \hookrightarrow SO(3) \xrightarrow{\pi} S^2$. The total space $SO(3)$ parameterises oriented tangent frames; π extracts the contact point; the $SO(2)$ fibre captures the tangent-plane orientation.

H Audit Table

I List of Assumptions, Limitations, and Conjectures

Assumptions

1. The candy is initially a perfect sphere of radius $R_0 \approx 12.30$ mm.
2. Hard-candy density $\rho = (1.54 \pm 0.06)$ g/cm³.
3. Equatorial radius preserved; only polar radius reduced.
4. Removal field is a sum of anisotropic Gaussian kernels.
5. Cutoff error is negligible for $\sigma \ll \pi$.
6. Kernel parameters $(\sigma, \rho_{\text{aniso}})$ are calibration parameters.

7. Inaccessible region is a spherical cap of colatitude θ_{stick} .
8. WGS84 flattening is small: first-order model is valid.
9. Total removed volume $\leq \Delta V = fV_0 \approx 26.1 \text{ mm}^3$.

Testable conjectures

1. For $\sigma = 0.1 \text{ rad}$, the minimum number of licks achieving $E_N < 10^{-3} \text{ mm}$ is $N_{\min} \in [3, 20]$.
2. For $\rho \geq 2$, Frame-SO3 achieves at least 20% smaller E_N than C-only for $N = 5$.
3. The latitude-circle holonomy at $\theta_0 = \pi/4$ is numerically $1.845 \pm 0.001 \text{ rad}$ (analytically $2\pi(1 - 1/\sqrt{2})$).
4. Sequence-order dependence in Experiment D: $\|\delta_N^{(A)} - \delta_N^{(B)}\|_{L^2} \propto (\rho - 1)$ for small $\rho - 1$.
5. Frank–Wolfe over measures on $SO(3)$ converges at $O(1/k)$.

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Table 4: Claim audit. Every major claim is classified.

Claim	Status	Evidence	Location	Remaining work
δ_{target} has only $\ell = 0, 2$ modes	Proved	Legendre algebra	Proposition 3.2	None
$F^+S^2 \cong SO(3)$	Std. cited result	Kobayashi–Nomizu	Proposition 5.2	None
$SO(3)/SO(2) \cong S^2$	Std. cited result	Lee, Ch. 21	Proposition 5.3	None
Isotropic kernel is $SO(2)$ -invariant	Proved	Direct computation	Proposition 6.5	None
Anisotropic kernel is orientation-sensitive	Proved	Explicit construction	Proposition 6.6	None
Holonomy $= \int K dA$	Std. cited result	Gauss–Bonnet	Theorem 7.2	None
Latitude-circle holonomy $2\pi(1 - \cos \theta_0)$	Proved (ODE)	Direct integration	Example 7.4	Numerical verification
Parallel transport rotates frame by $\Delta\psi$	Proved	Theorem 7.2	Proposition 7.5	None
Sequence-order dependence (anisotropic)	Proved (construction)	Triangle holonomy $\pi/2$	Proposition 7.6	Numerical example
Isotropic order invariance	Proved	Commutativity of sum	Proposition 7.7	None
OMP finds global min lick count	NOT claimed	NP-hardness	Algorithm 2	N/A
vMF is globally stationary Gaussian	NOT claimed	Position-dependent width	Definition 4.3 (remark)	N/A
Stereo. Fourier = global isomorphism	NOT claimed	Conformal factor	Section 15	N/A
Tangent planes form a group	NOT claimed	Correct: frame bundle	Remark 5.4	N/A
Optimality in $a(t)$ (bang-bang)	Proposed/derived	PMP sketch	Proposition 9.4	Regularity verification
Adjoint frame equation	Proposed	Lie-group PMP analogy	Proposition 9.5	Full derivation
Minimum lick count scaling	Conjectured	CS analogy	Conjecture 12.1	Proof or experiment
$SO(3)$ strategy outperforms S^2	Conjectured	Conjecture 15.2	Section 11	Numerical experiments A–F
Frank–Wolfe convergence on $SO(3)$	Conjectured	Chizat–Bach analogy	Conjecture 15.3	Proof